DEEP NETWORK DEVELOPMENT

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Deep Network Development

Lecture 3.

Image Classification & Convolutional Neural Networks

Budapest, 1st March 2024

1 **Image Classification**

2 **Convolutional Neural Networks**

Assignments

- 2 Assignments. All assignments must be done using PyTorch
- All assignments have a defense consisting of two parts:
  - Code part: instructor asks questions related to the code of your submission
  - Theoretical part: instructor asks questions related to theoretical aspects of the topic of the assignment

- All assignments will be released next week (Wednesday or Thursday, March 6th /7th). During next lecture I will go through the details of the assignments.
- All the details, guidelines, evaluation criteria and defense topics will be available on Canvas

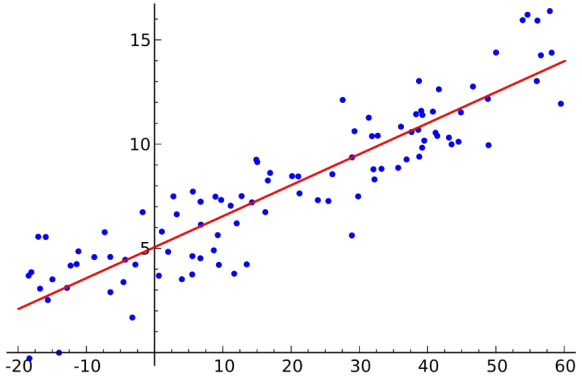
- **Assignment 1: Object Detection**
  - **Deadline: 17th April (approx. 6 weeks from release date)**

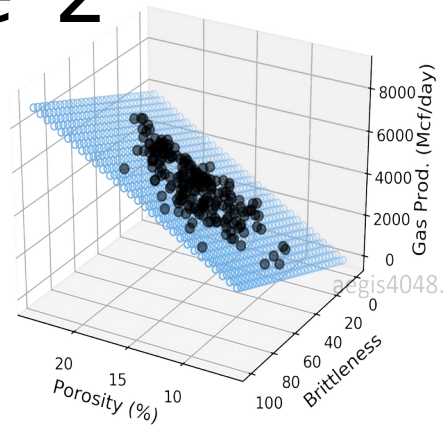
- **Assignment 2: Multimodal Language Model (Vision + Language)**
  - **Deadline: 15th May (approx. 10 weeks from release date)**

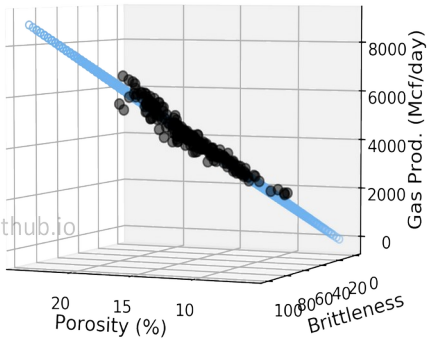
- You can contact the demonstrators at any time for consultation. **Deadlines are**

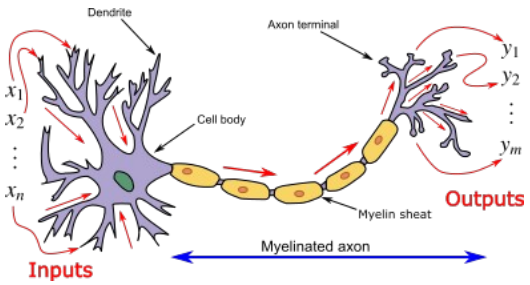
Previously on Lecture 2

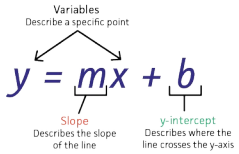
Use Linear Regression to predict Life Expectancy



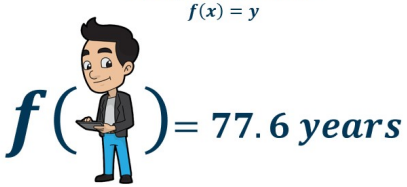




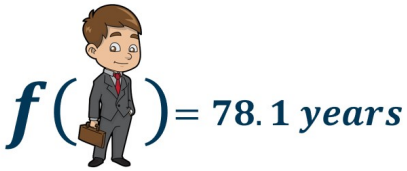




Life Expectancy function
$f(x) = y$



$f(\quad) = 77.6$ *years*



$f(\quad) = 78.1$ *years*

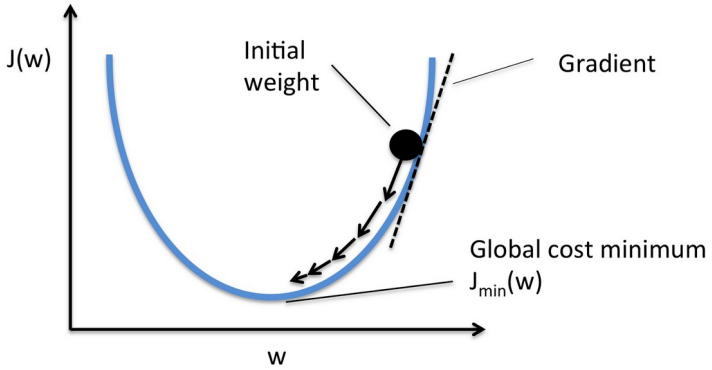
$$\hat{y} = \boldsymbol{\theta_1 x + \theta_0} \quad \hat{\mathbf{Y}} =$$

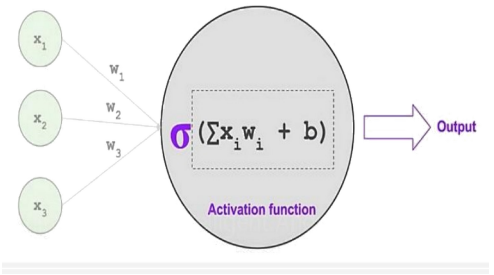
$$\boldsymbol{\theta_1, \theta_0} = ?$$

$$J(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(\theta 1 x^{(i)} + \theta 0 - y^{(i)}\right)^2$$

$$\boldsymbol{J(\theta) = (X\theta - Y)^T (X\theta - Y)}$$

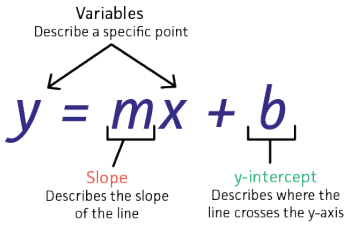
Normal Equation:

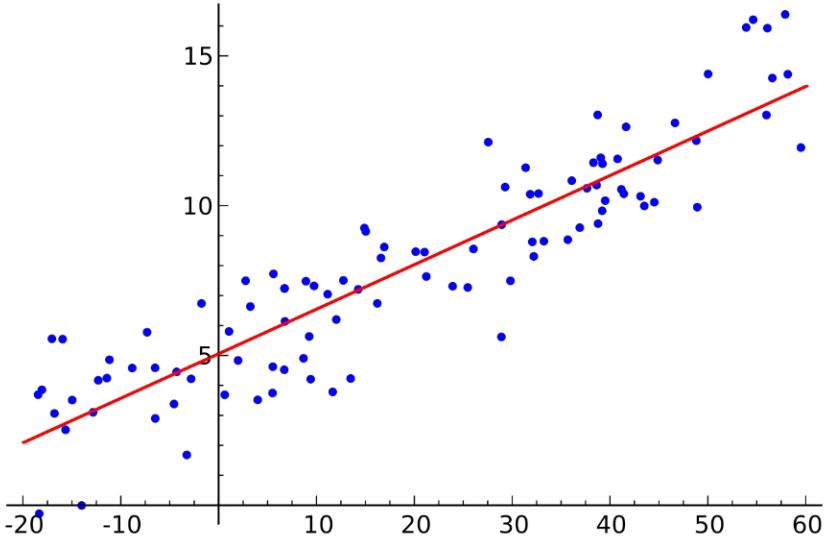




Linear Regression

- Supervised learning
- Have: (x, y)
  - x – input
  - y – target
- *Goal: Learn a function to map* $x \rightarrow y$.



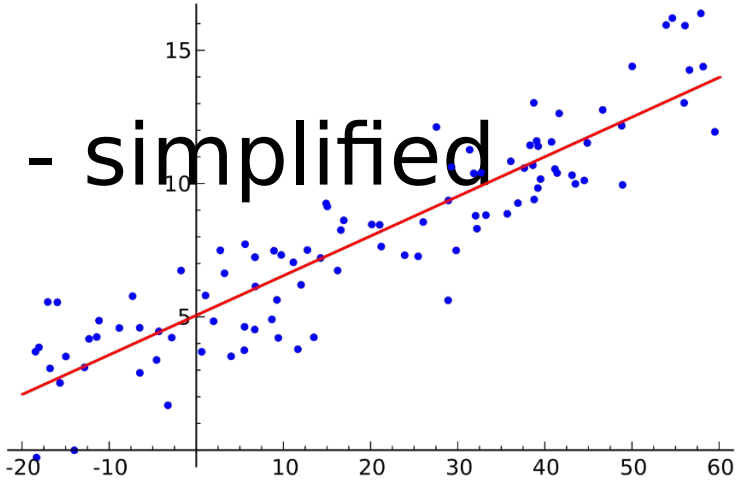


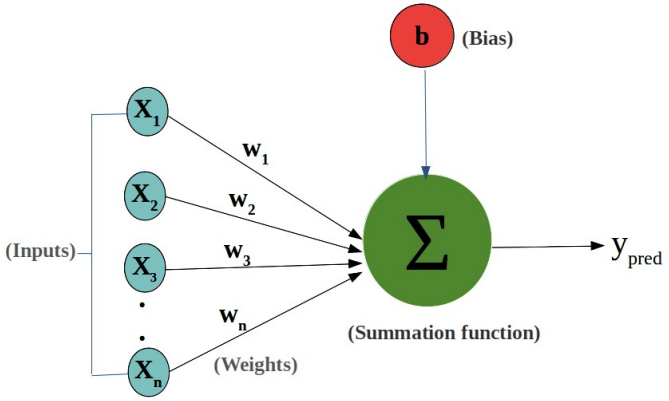
- Regression – Predict real valued / continuous output:
  - $\in \mathbb{R}$

- Given the height and weight of a person, predict the age of the person.
- Age can be: 10,25,33,71,...

Artificial Neural Network (ANN) - simplified

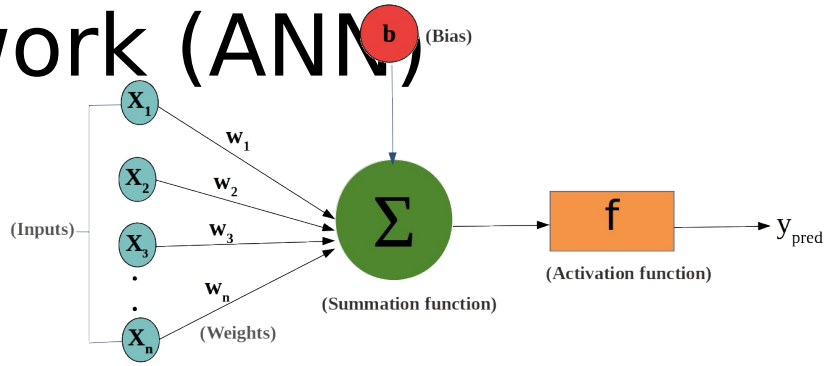
- Using ANN for linear regression
- – the inputs
- – weight parameters
- *b* – bias parameter
- *o* – the output
- One neuron with *m* inputs does the following:

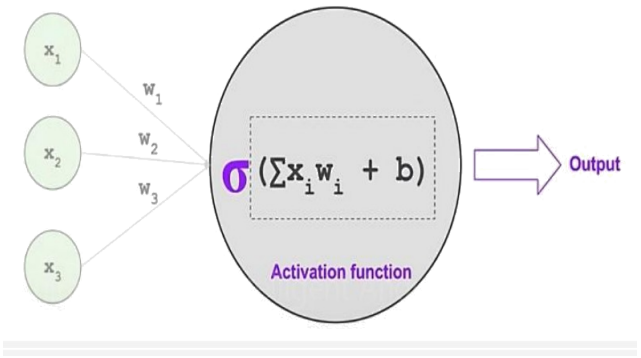




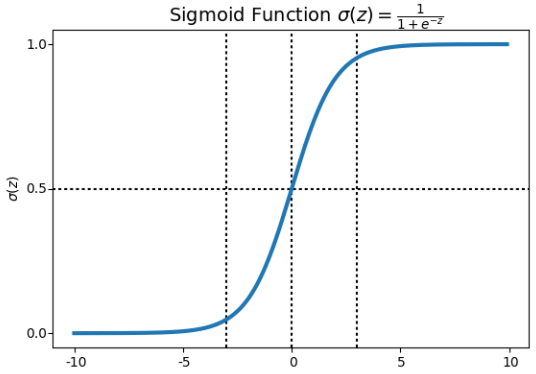
- Given the height and weight of a person, predict the age of the person.
- Age can be: 10,25,33,71,...

Artificial Neural Network (ANN)





- Using ANN for linear regression
- – the inputs
- – parameters we will train
- *b* – bias parameter
- ***g* – nonlinear activation function**
- *o* – the output
- One neuron with *m* inputs does the following:



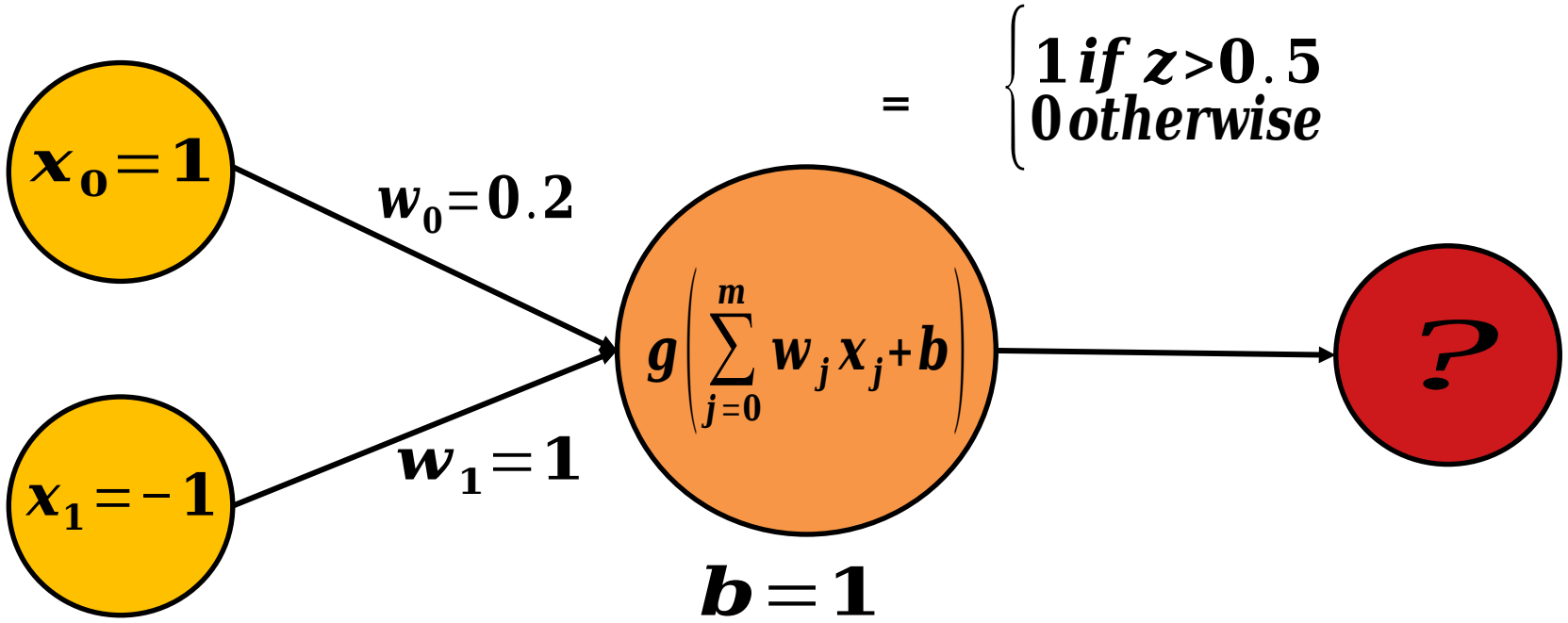
$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Activation function

- Adds non-linearity
- Output limited to a range (i.e. 0-1)
- Given the height and weight information, predict if it is a human or an animal.
- 0: human; 1: anim

Question 1

What is the output value of this neuron?



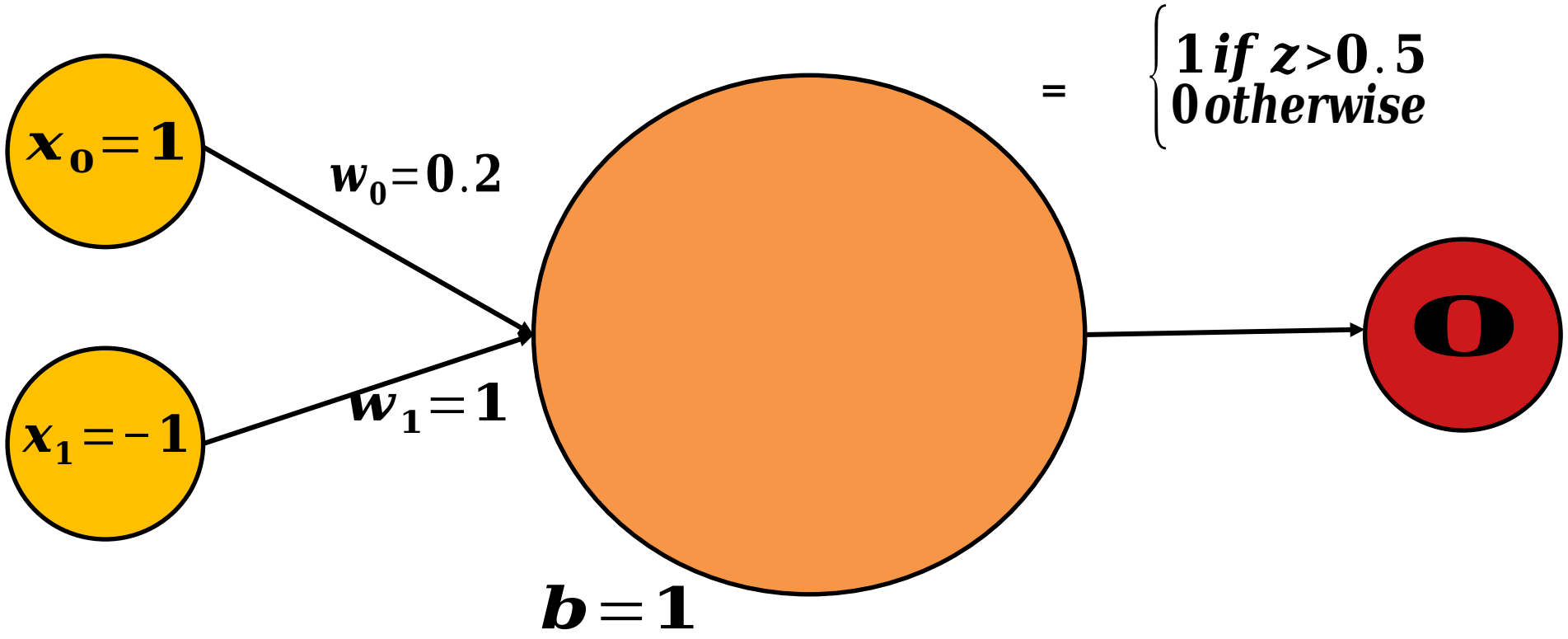
Answers

a) 1

b) 0

Question 1 - solution

What is the output value of this neuron?



Answers

a) 1

b) 0

Deep Network Development

Lecture 3.

Image Classification & Convolutional Neural Networks

Budapest, 1st March 2024

1 Image Classification

2 Convolutional Neural Networks

Linear Regression vs Classification

Linear Regression

- Supervised learning
- Have: (x, y)
  - x: input
  - y: target $\in \mathbb{R}$
- *Goal: Learn a function to map $x \rightarrow y$.*
- Regression – Predict real valued / continuous output:
  - $\in \mathbb{R}$

Binary Classification

- Supervised learning
- Have: (x, y)
  - x: input
  - y: target $\in \{0, 1\}$
- *Goal: Learn a function to map $x \rightarrow y$.*
  - Add non-linear activation function
- Classification – Predict discrete set of values:
  - $\in \{0, 1\}$

Binary Classification

Binary classification – which of the two classes does the input belong to?

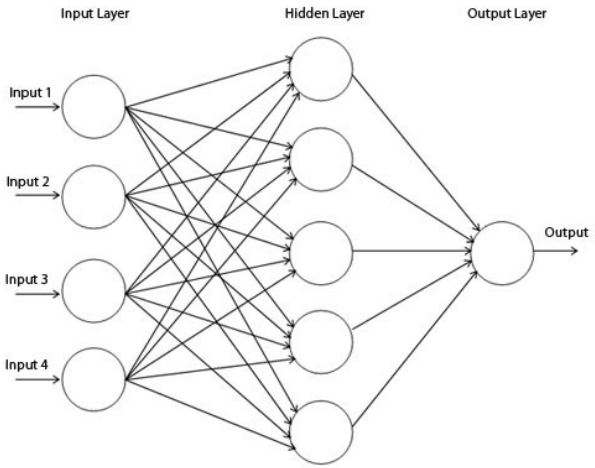
- Cat vs Dog
- Dog vs Mop

Binary Classification

Binary Classification

- Binary classification – Two classes
- Is x in the target class C?
- Output $P(x \in C)$, the probability of x is in C
- The network will have 1 output
- h(x) must be between 0 and 1

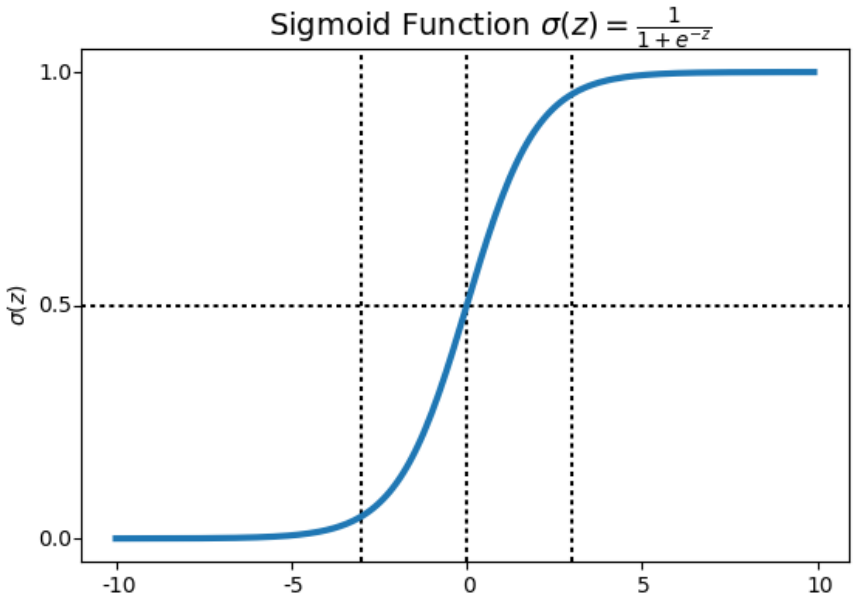
Dog vs Mop



Binary Classification

- h(x) must be between 0 and 1
- Use the sigmoid activation on last layer

$$\sigma(z) = \frac{1}{1+e^{-z}}$$



- Linear regression becomes Logistic Regression (Binary Classification), if we add a Sigmoid activation function to the output
- Linear Regression: (continuous output)
- Binary Classification:

Binary Classification Loss

Linear Regression

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

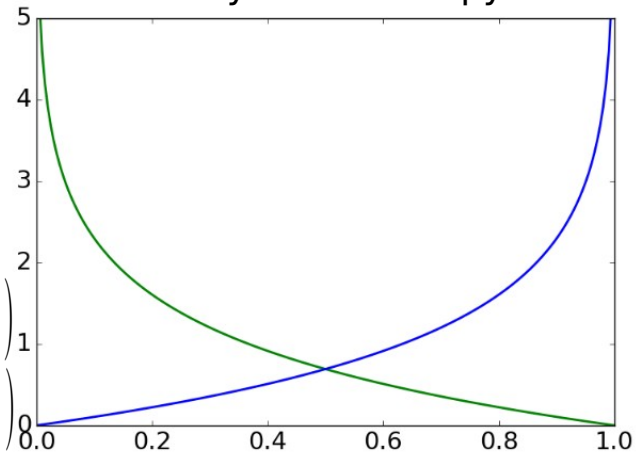
- Binary cross-entropy loss

- Assumptions:
  y is 0 or 1
  $h(x) = \hat{y}$ is between 0 and 1

Binary cross-entropy

Loss

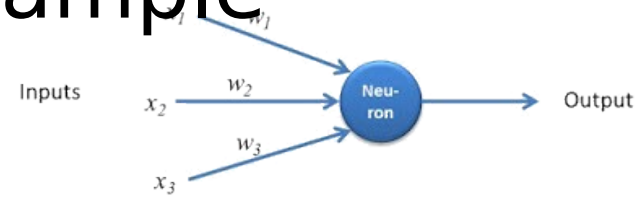
$$L_{BC}(h(x),1)$$
$$L_{BC}(h(x),0)$$

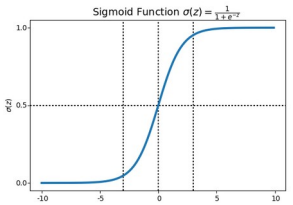


Prediction

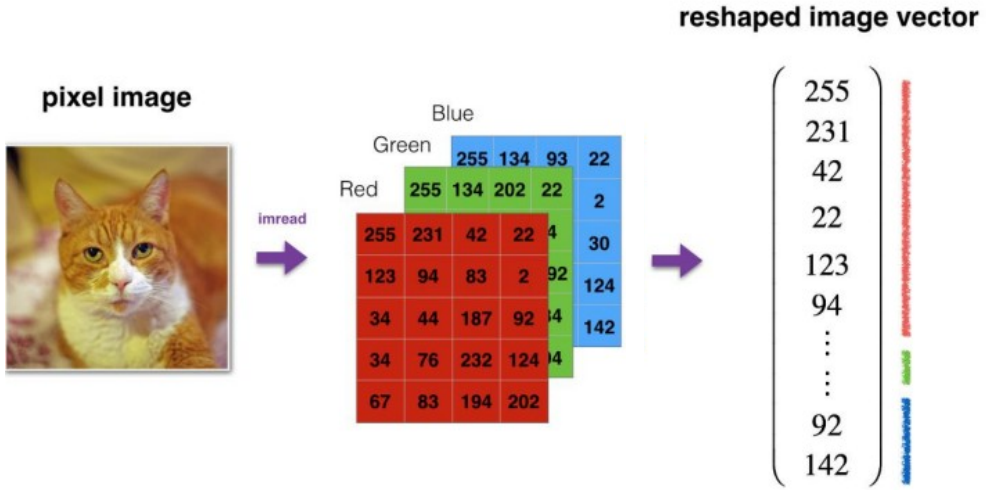
Binary Classification example

Classes C = {0: Dog, 1: Cat}

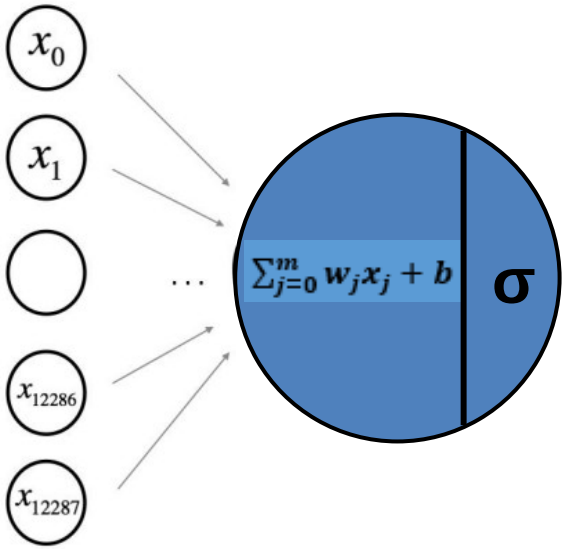




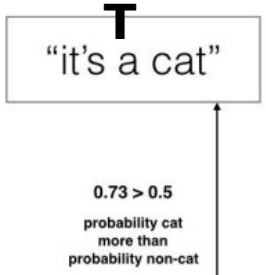
INPUT



NEURAL NETWORK



OUTPUT



Sigmoid Activation Function → 0.73

TARGET

Y = 1

LOSS

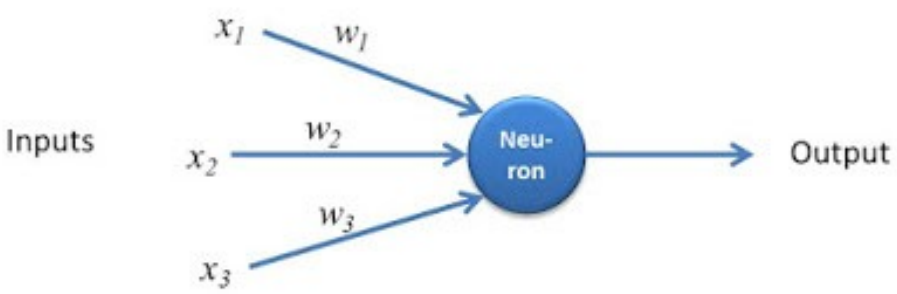
More neurons and categories

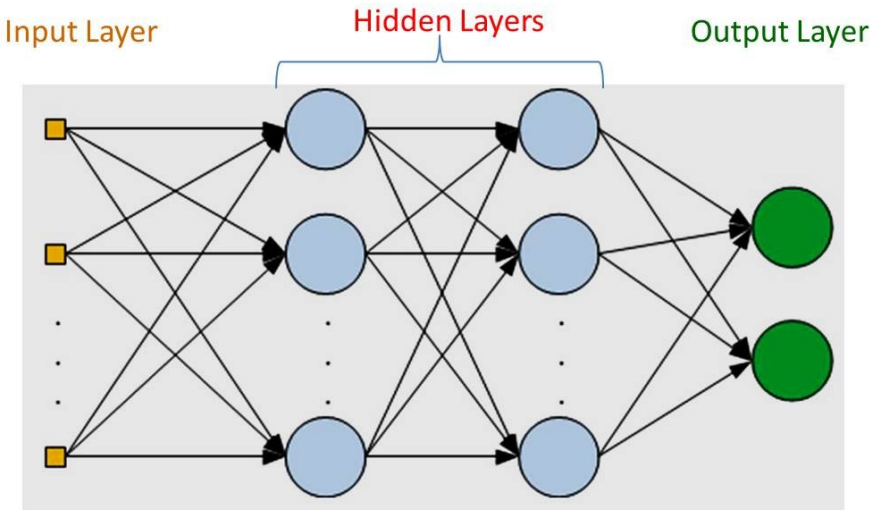
Next

- Single artificial neuron → Deep Neural Network
- Binary Classification (0 or 1) → Multi-Class Classification (0,1,2,3…)

Deep Neural Networks

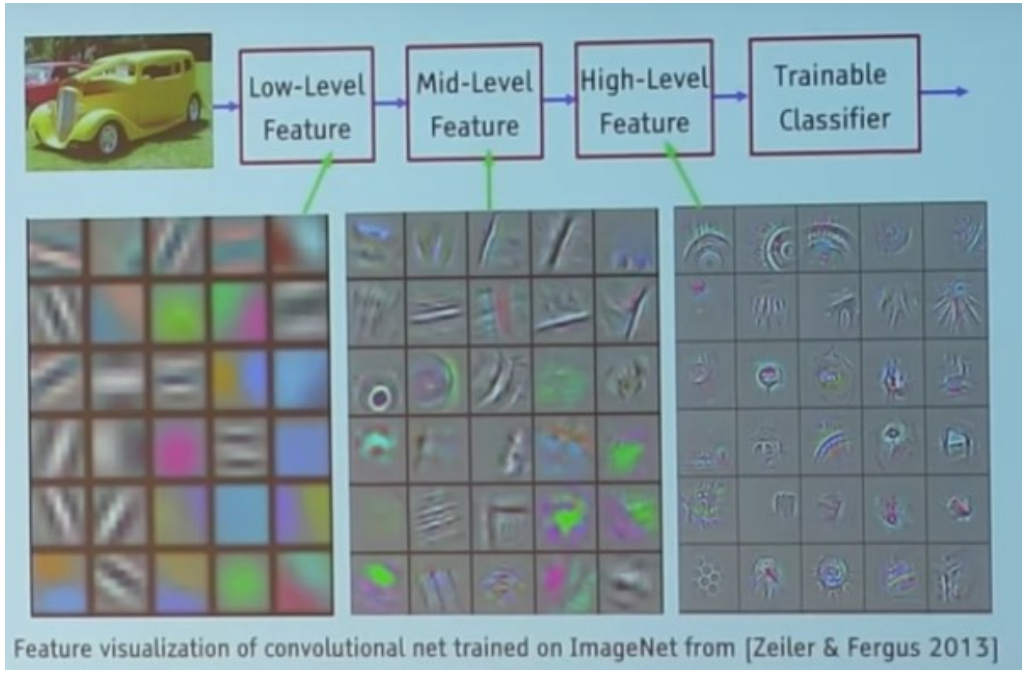
- Neural networks are built up from neurons, that have inputs and outputs
- Neurons are organised into layers
- Layers refine the output of the previous layers



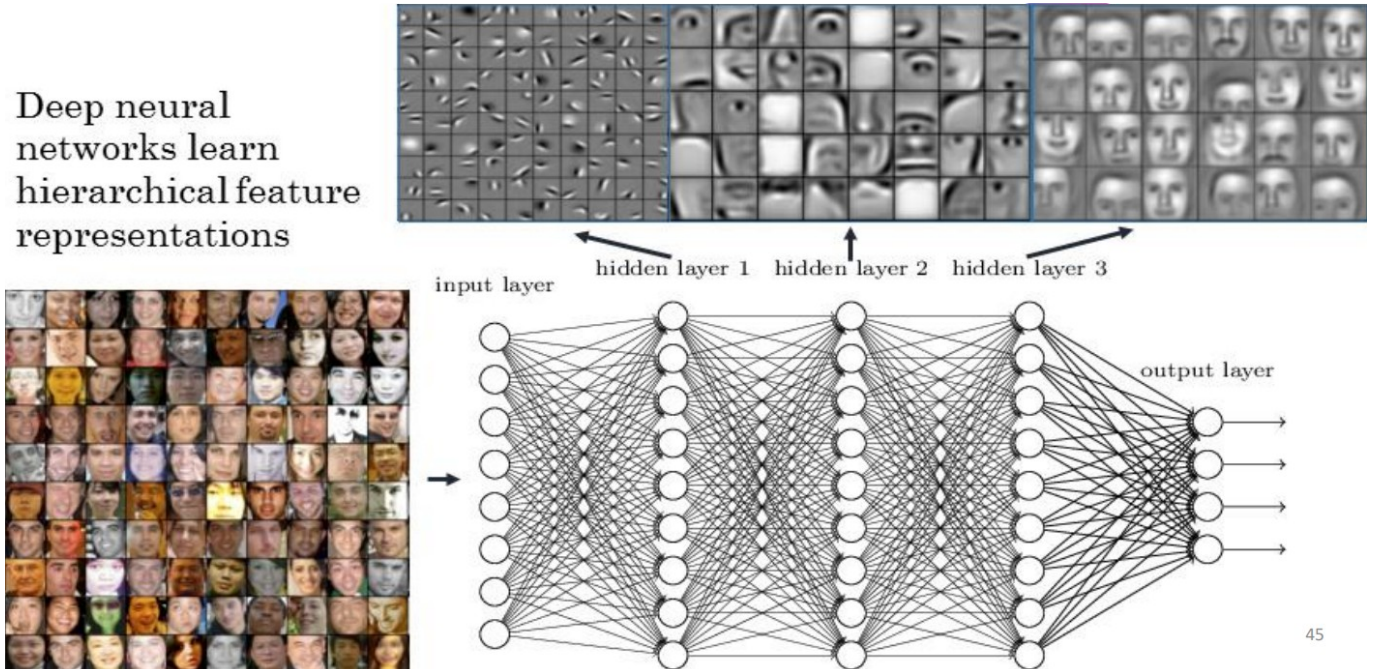


Deep Neural Networks

- Each layer refines the previous layer
- Visualization demo: https://adamharley.com/nn_vis/



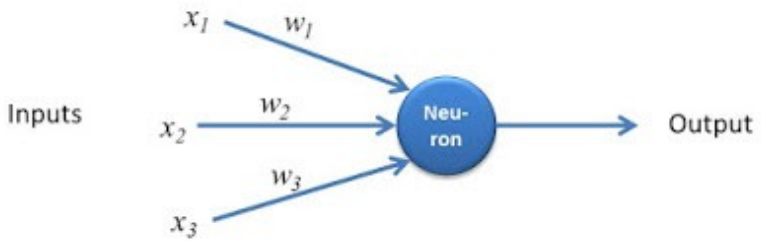
Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

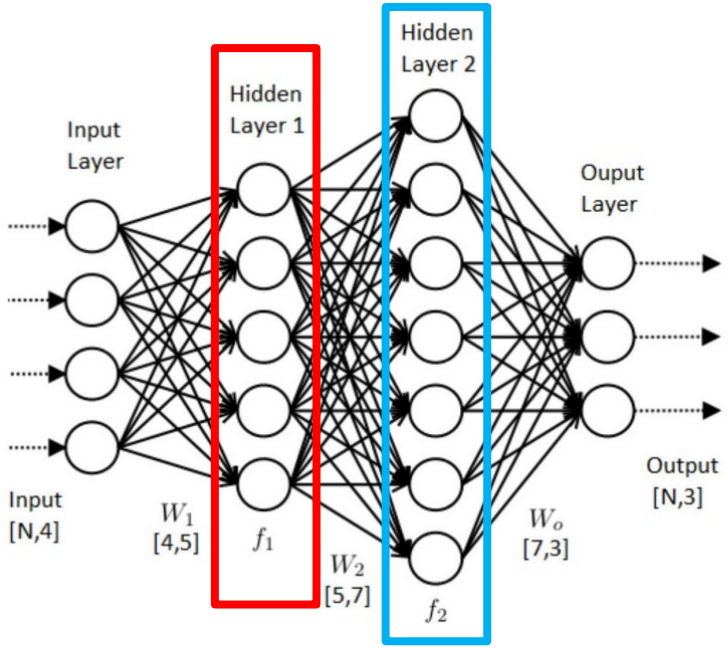


Deep Neural Networks

- Artificial Neural Networks ⟺ Feed Forward Neural Networks ⟺ Fully Connected Networks

- A neural network is built up from neurons
- Neurons are organised into layers
- Layers refine the output of the previous layers





Deep Neural Networks

- How to find *W* and *b*?
- Given

- Select a loss function L that measures how good *h* is:
  - the smaller L(h(x),y) , the better *h* is

- Update :

- Update :

- Update :

- Update :

Terminology

- L(h(x),y) – loss or error function for a single data point of the dataset
- C(θ), J(W) – cost or objective function for the entire dataset (usually average)
- The update rule can also be in the form:

$$w_i \leftarrow w_i + \Delta w_i$$

$$b \leftarrow b + \Delta b$$

$$\Delta w_i = -\alpha \frac{\partial L}{\partial w_i}$$

$$\Delta b = -\alpha \frac{\partial L}{\partial b}$$

[1] Backpropagation derivation: https://www.cs.put.poznan.pl/pliskowski/pub/teaching/eio/lab1/eio-supplementary.pdf
[2] Another backpropagation derivation (be aware of notation differences): https://www.cs.swarthmore.edu/~meeden/cs81/s10/BackPropDeriv.pdf

Deep Neural Networks

1. Select randomly
2. Calculate for all training examples
3. Calculate the cost function:

4. Update W:

5. Repeat steps 2-4 until convergence

Deep Neural Networks

1. Select W_0 randomly
2. Calculate $h(X)$ for all training examples
3. Calculate the cost function:

$$J(W) = \frac{1}{N}\sum_{i=1}^{N} L(h(x_i), y_i)$$

4. Update W:

$$W_t = W_{(t-1)} - \alpha \nabla J(W)$$

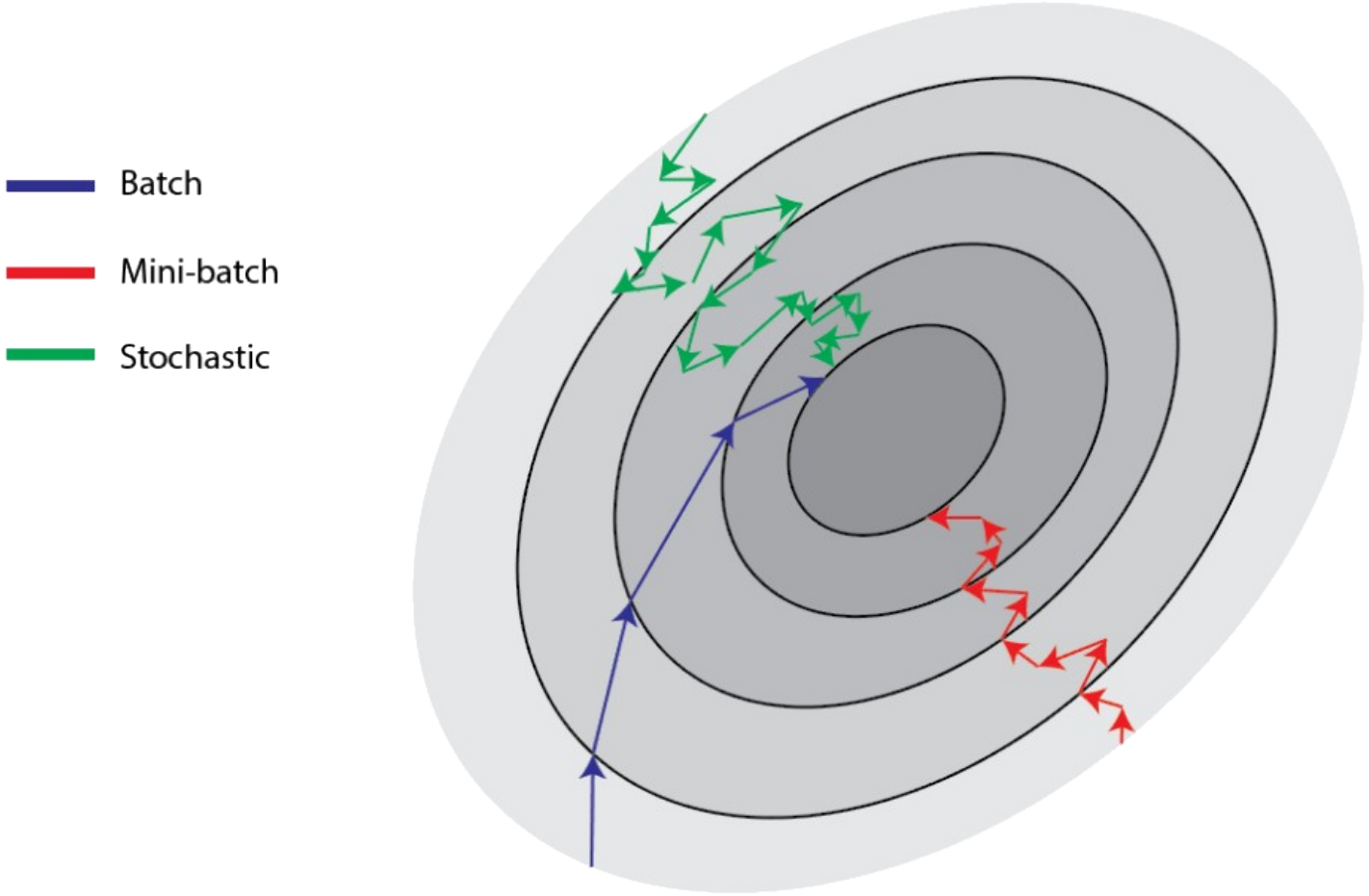
5. Repeat steps 2-4 until convergence

Batch size – the number of training samples used to optimize the model's parameters

Types (with Gradient Descent)

- **Batch Gradient Descent**
  - Batch size = Size of dataset
- **Mini-Batch Gradient Descent**
  - Batch size = a subset of the dataset examples; typically, 2^n , e.g., 8,16,32,128,...
  - In each iteration, a mini-batch is randomly sampled from the dataset
- **Stochastic Gradient Descent**
  - Batch size = 1
  - In each iteration, a single example is randomly sampled from the dataset

Training Deep Neural Networks



Hyperparameter vs Parameter

Hyperparameters

To create and train Neural Networks we must make certain decisions (we choose the values):

- Number of layers; Number of neurons in each layer
- Activation functions
- Learning rate
- Batch size
- Many more (optimization related)

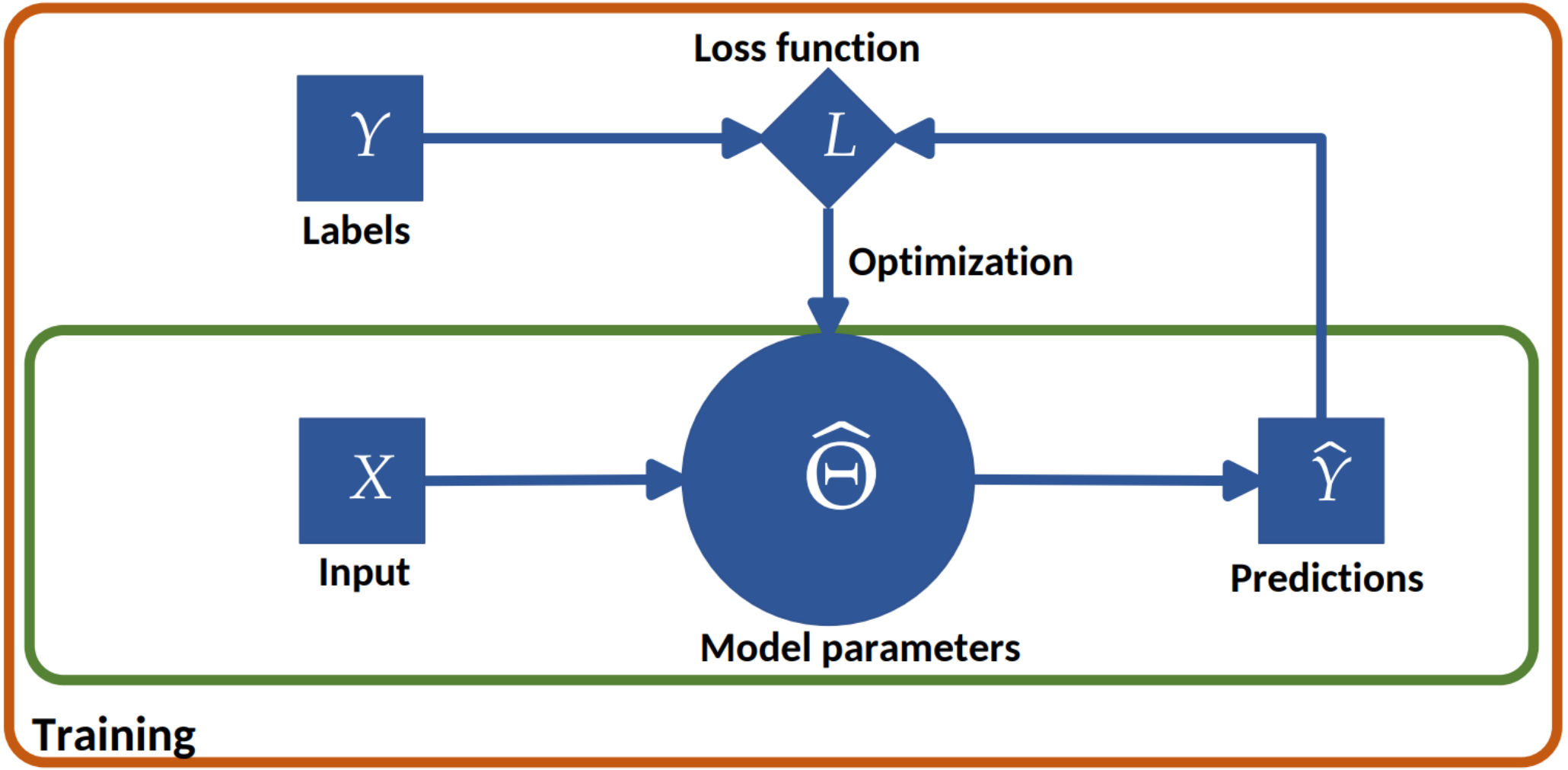
To be continued in next lecture…

Parameters

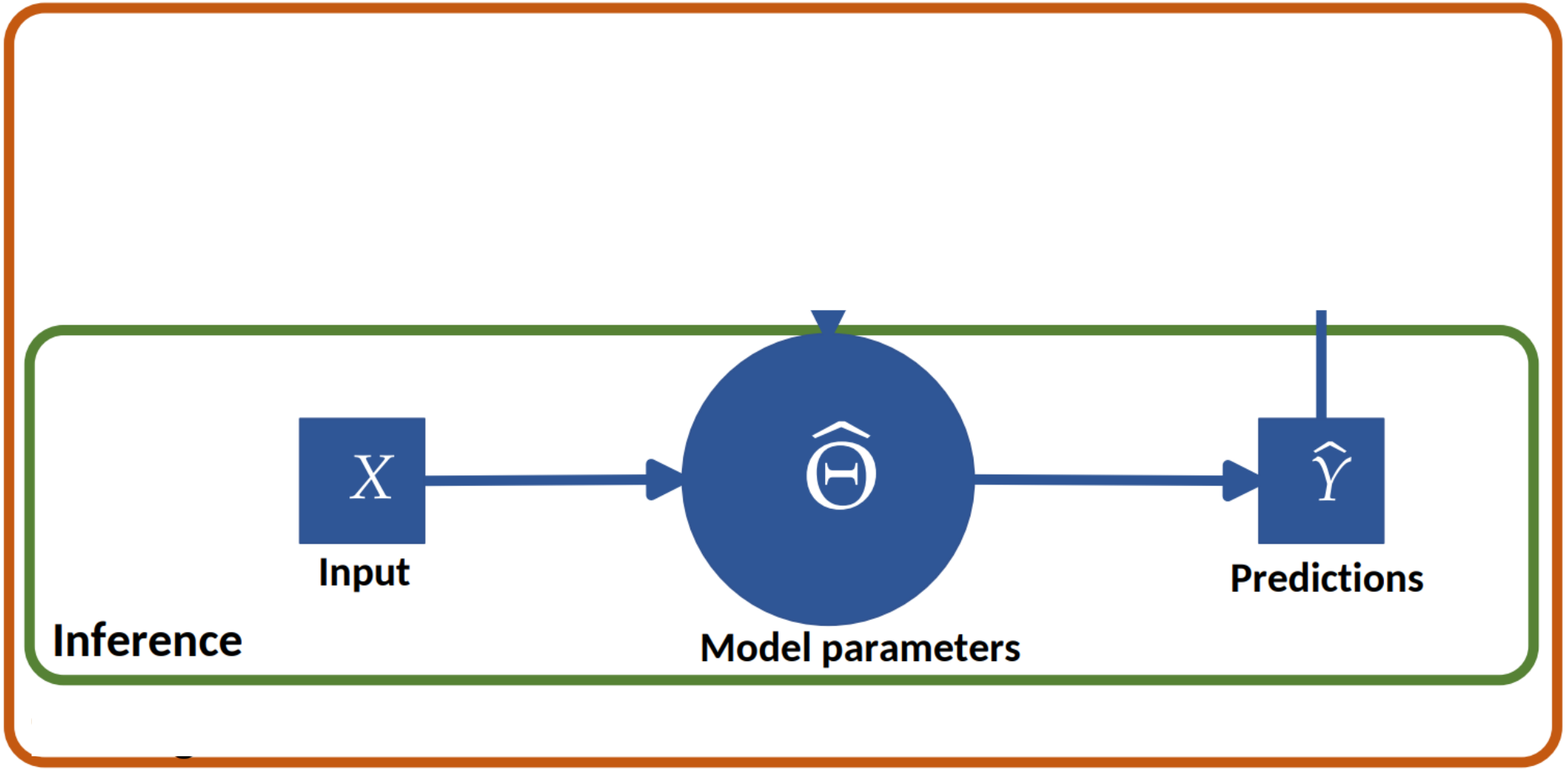
What the Neural Network learns. The weights and biases of the network are optimized with an algorithm (Gradient Descent, etc.). We do not choose the values for them.[3]

3. We can choose the initial values (weight initialization), it can be initialized with zeros, ones, random values, etc. However, the network learns the optimal values through training.

Deep Neural Networks



Deep Neural Networks



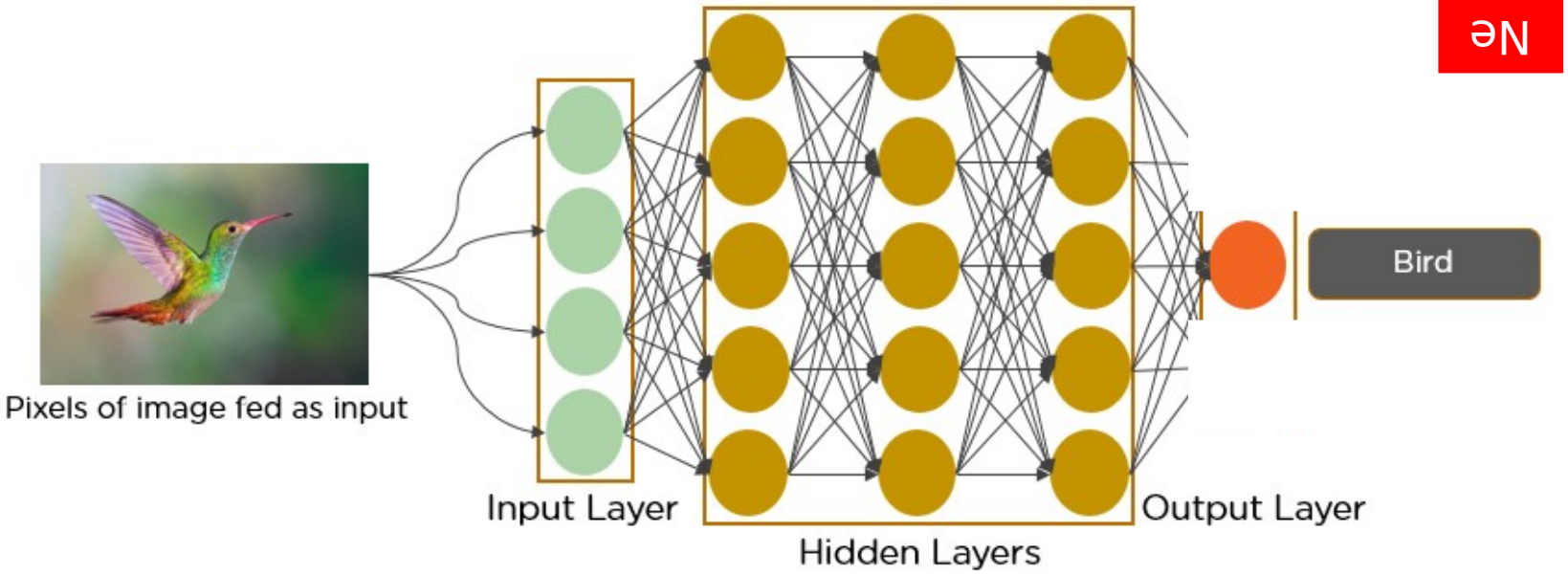
More neurons and categories

- **Single artificial neuron**
- Binary Classification (0 or 1)

→

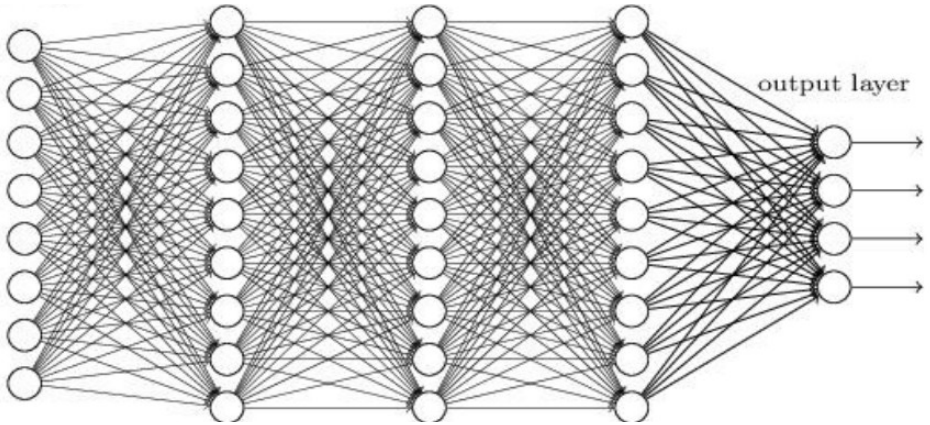
- **Deep Neural Network**
- Multi-Class Classification (0,1,2,3…)

Next



Multi-Class Classification

- With *K* classes
- The network will have K outputs
- Softmax activation for squashing outputs between 0 and 1



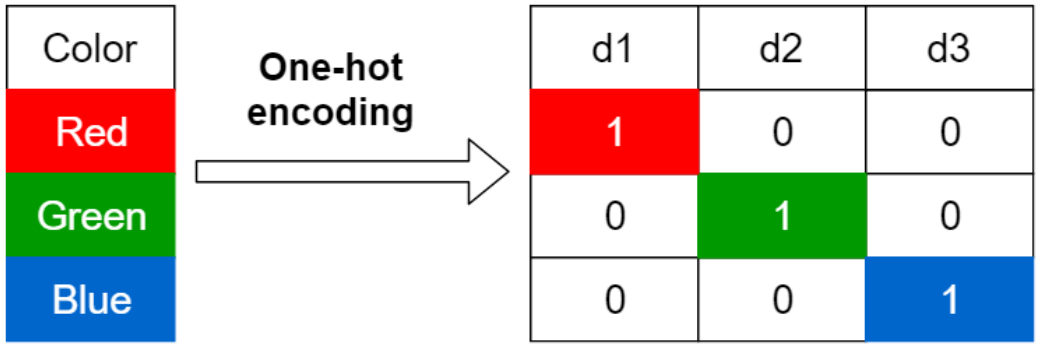
This is basically a probability distribution on the classes.
They sum up to 1.

Multi-class Classification

Output of the neural network is a K long vector

How should we encode the ground-truth values?
One-hot encoding (K=3):
(cat) 1 -> [1, 0, 0]
(dog) 2 -> [0, 1, 0]
(horse) 3 -> [0, 0, 1,]

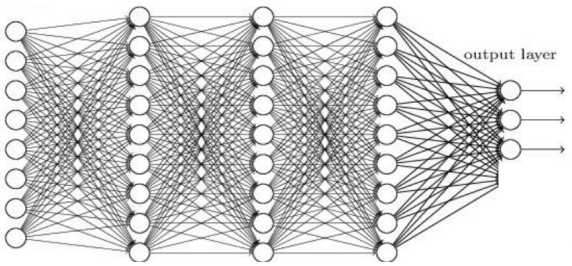
Just as h(x): values between 0 and 1, sum up to 1

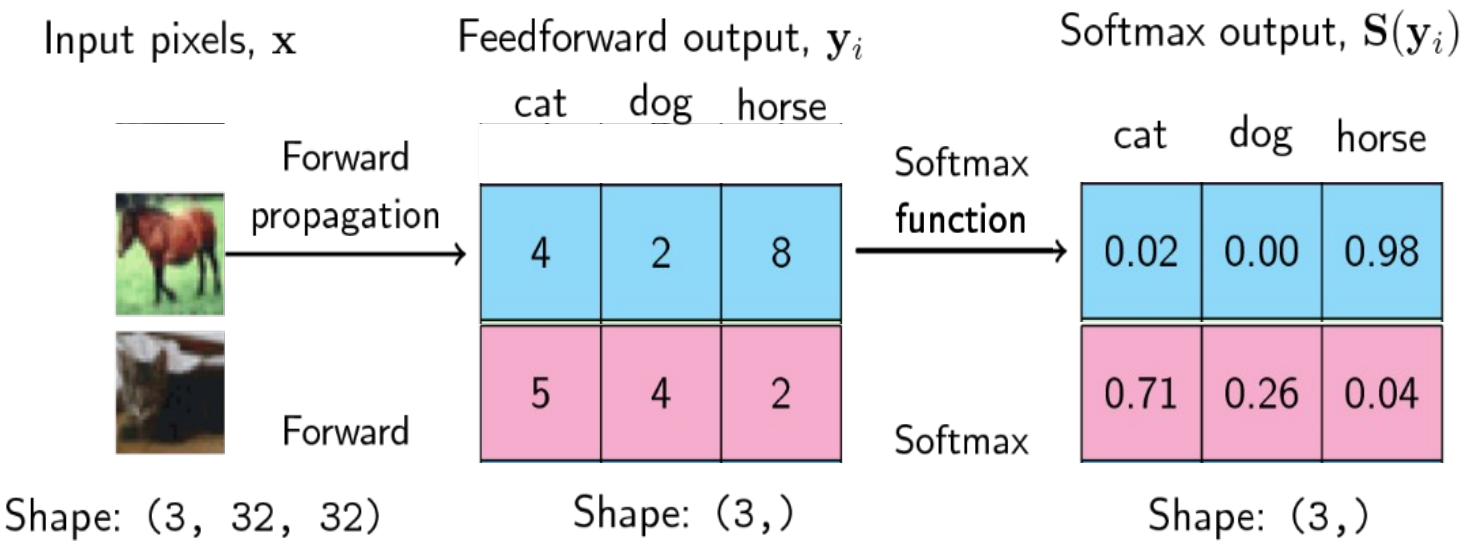


Multi-class Classification

- K = 3
- Softmax activation
- Probability distribution on the classes. They sum up to 1.

(Cat, Dog, Horse)





Multi-class Classification Loss

With K classes
Categorical Cross-entropy loss

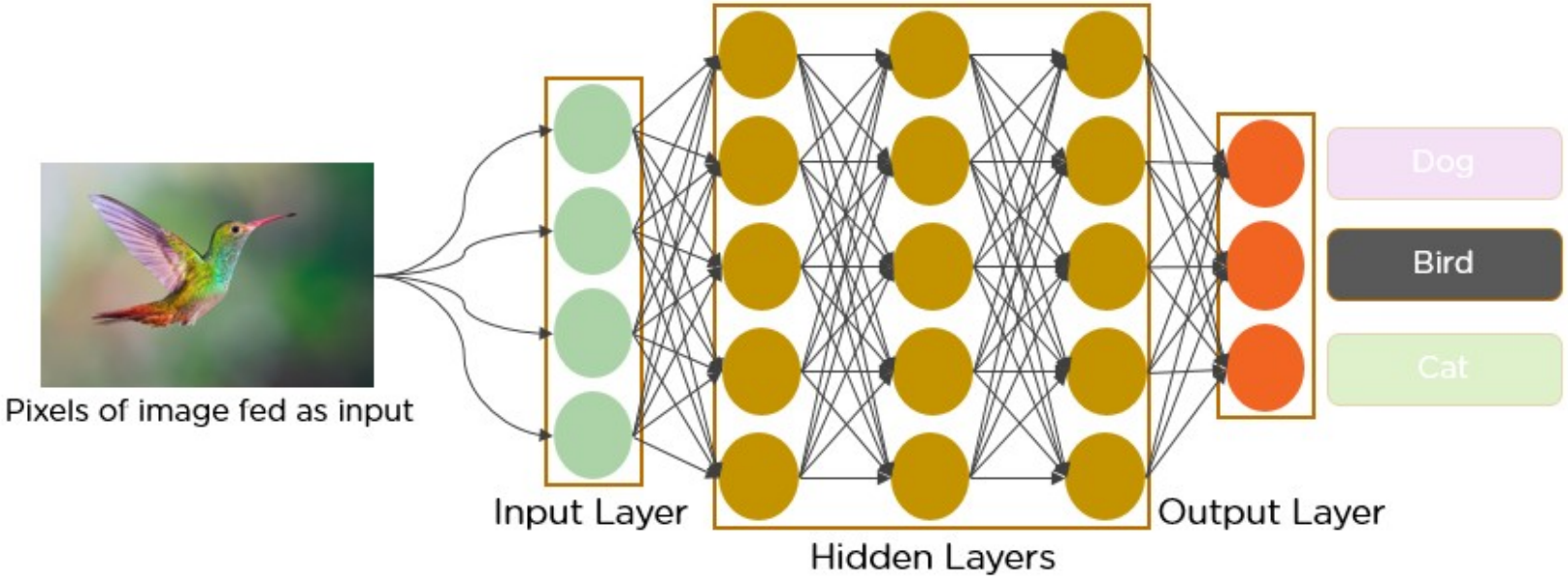
Assumptions:

y is a vector of K *elements with values* 0 or 1, exactly one element is 1

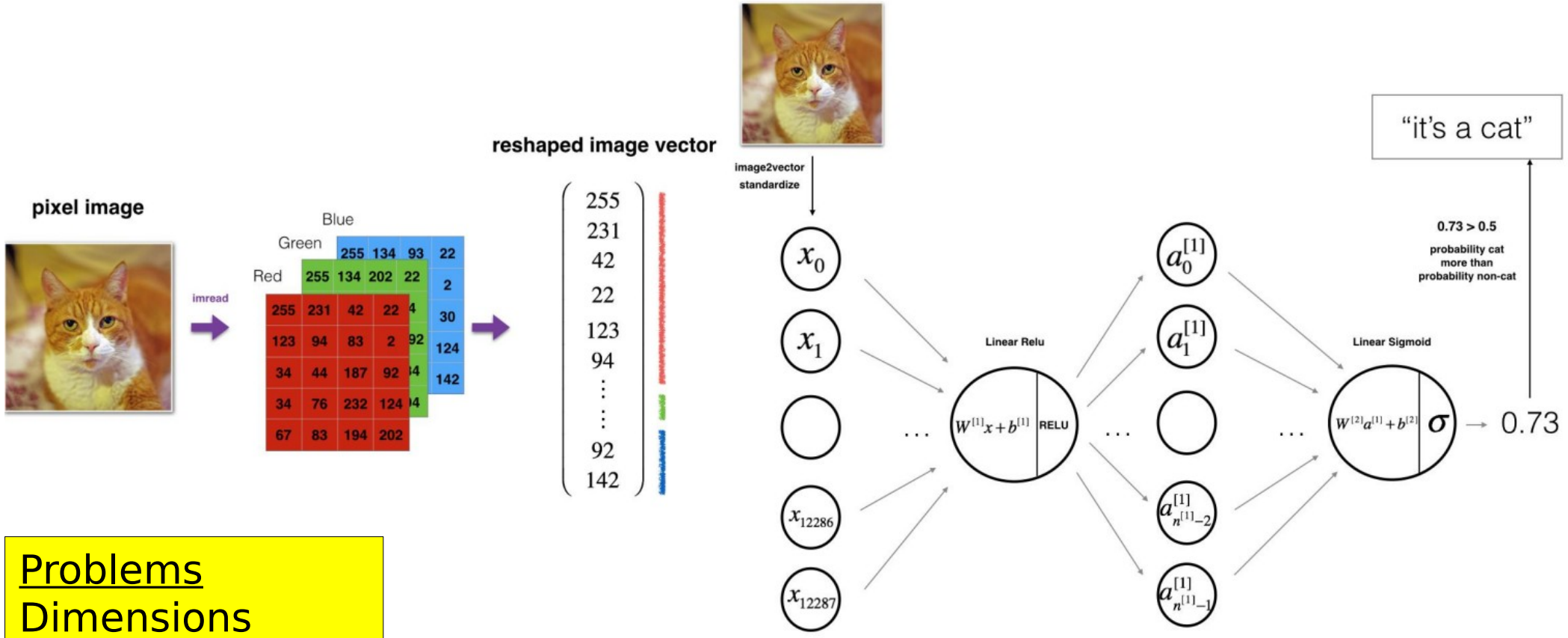
$h_i(x)$ is a vector of K elements between 0 and 1, the sum of all elements is equal to 1

More neurons and categories

- **Single artificial neuron** → **Deep Neural Network**
- **Binary Classification (0 or 1)** → **Multi-Class Classification (0,1,2,3…)**



Disadvantages of Feed Forward Neural Networks

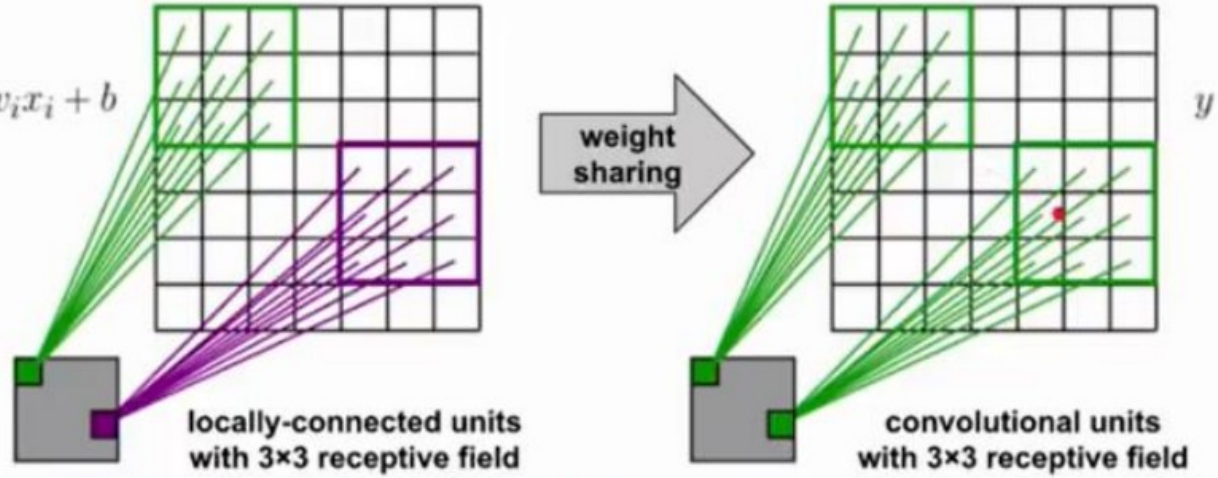


<u>Problems</u>
Dimensions
Weight Sharing

Disadvantages of Feed Forward Neural Networks (Intro to CNNs)

- **Weight sharing**
  - units connected to different locations have the same weights
  - equivalently, each unit is applied to all locations
- *Convolutional layer – locally-connected layer with weight sharing (translation invariance)*
- The weights are invariant, the output is equivariant

$$y = \sum_{i \in \text{receptive field}} w_i x_i + b$$



$$y = w * x + b$$

Practice: Architectures : Convolutional Nets

Deep Network Development

Lecture 3.

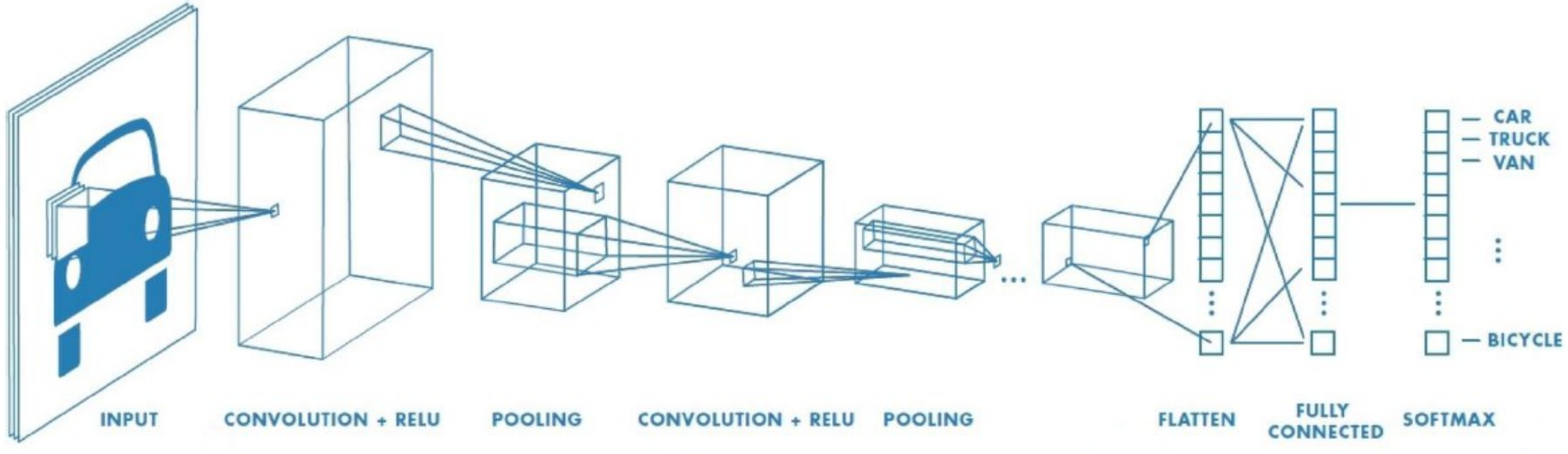
Image Classification & Convolutional Neural Networks

Budapest, 1st March 2024

1 **Image Classification**

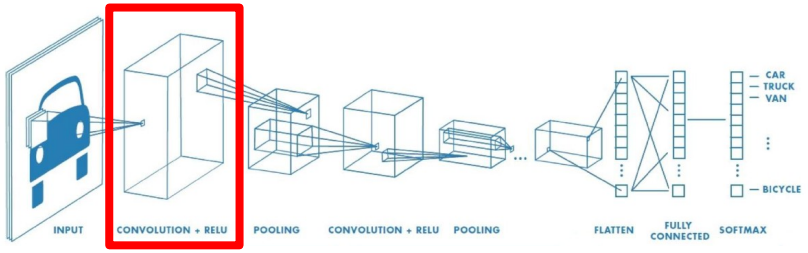
2 **Convolutional Neural Networks**

How do Convolutional Neural Networks (CNNs) work?



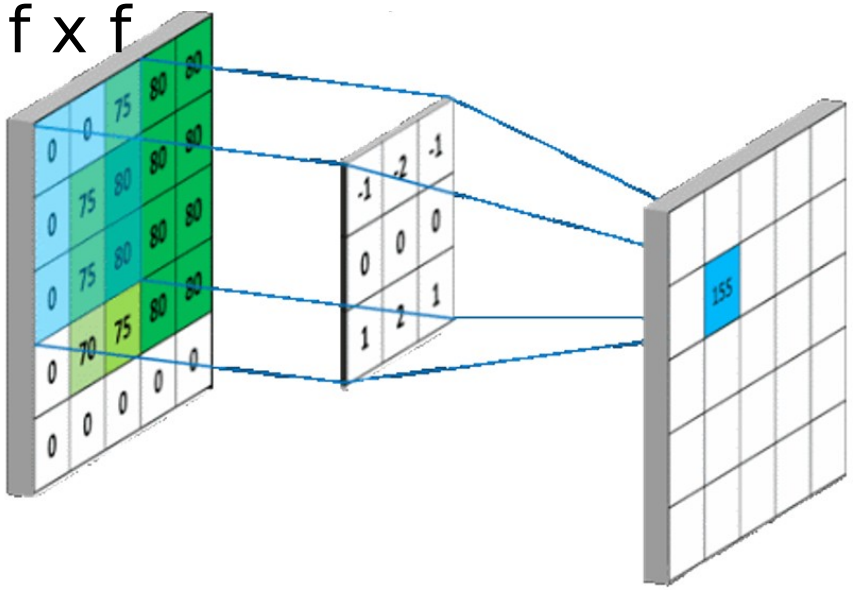
CNNs extract relevant features from an image.
Like a magnifying glass “enhancing” details of the image.

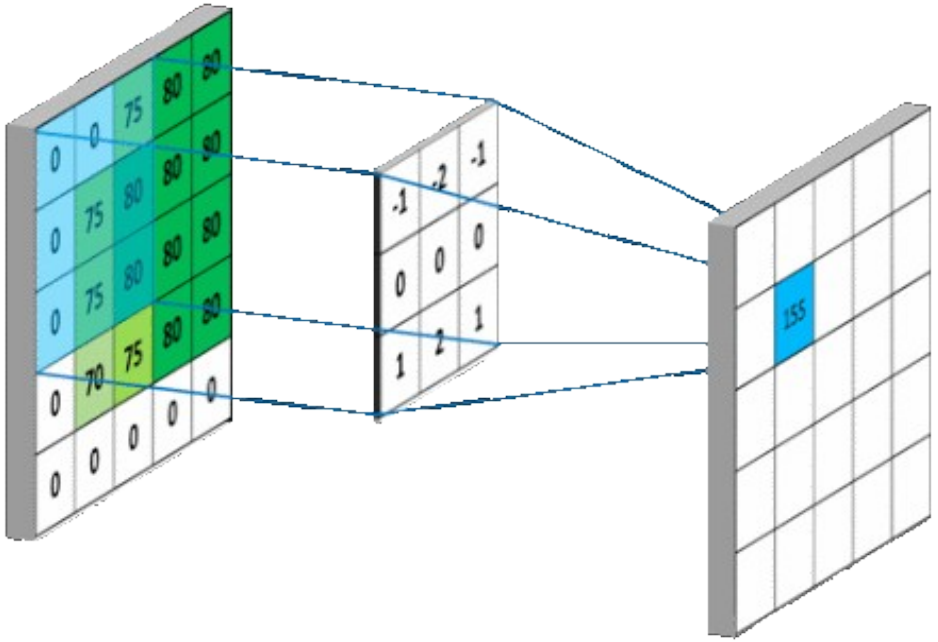
Convolutional layer



Convolution
Input: 5x5
Filter: 3x3
Output: 3x3

Filter / Kernel size =
f x f





Convolutional layer

- Convolution operation
- Input: 5 x 5

10	2	4	1	1
0	4	0	2	2
1	8	16	0	0
8	1	1	0	0
0	0	2	6	1

*

- Filter / kernel size = 3 x 3

2	2	2
1	0	1
2	2	2

- Output = ?

Convolutional layer

- Convolution operation
- Input: 5 x 5
- Filter / kernel size = 3

10 x 2	**2** x 2	**4** x 2	**1**	**1**
0 x 1	**4** x 0	**0** x 1	**2**	**2**
1 x 2	**8** x 2	**16** x 2	**0**	**0**
8	**1**	**1**	**0**	**0**
0	**0**	**2**	**6**	**1**

10 x 2 + 2 x 2 + 4 x 2 +
0 x 1 + 4 x 0 + 0 x 1 +
1 x 2 + 8 x 2 + 16 x 2 = 82

- Output = ?

82		

Convolutional layer

- Convolution operation
- Input: 5 x 5
- Filter / kernel size = 3 x 3

10	**2** x 2	**4** x 2	**1** x 2	**1**
0	**4** x 1	**0** x 0	**2** x 1	**2**
1	**8** x 2	**16** x 2	**0** x 2	**0**
8	**1**	**1**	**0**	**0**
0	**0**	**2**	**6**	**1**

2 x 2 + 4 x 2 + 1 x 2
+
4 x 1 + 0 x 0 + 2 x 1
+
8 x 2 + 16 x 2 + 0 x 2
= 68

- Output = ?

82	**68**	

Convolutional layer

- Convolution operation
- Input: 5 x 5
- Filter / kernel size = 3 x 3

10	**2**	**4** x 2	**1** x 2	**1** x 2
0	**4**	**0** x 1	**2** x 0	**2** x 1
1	**8**	**16** x 2	**0** x 2	**0** x 2
8	**1**	**1**	**0**	**0**
0	**0**	**2**	**6**	**1**

4 x 2 + 1 x 2 + 1 x 2
+
0 x 1 + 2 x 0 + 2 x 1
+
16 x 2 + 0 x 2 + 0 x 2
= 46

- Output = ?

82	**68**	**46**

Convolutional layer

- Convolution operation
- Input: 5 x 5
- Filter / kernel size = 3 x 3

10	**2**	**4**	**1**	**1**
0 x 2	**4** x 2	**0** x 2	**2**	**2**
1 x 1	**8** x 0	**16** x 1	**0**	**0**
8 x 2	**1** x 2	**1** x 2	**0**	**0**
0	**0**	**2**	**6**	**1**

0 x 2 + 4 x 2 + 0 x 2
+
1 x 1 + 8 x 0 + 16 x 1
+
8 x 2 + 1 x 2 + 1 x 2
= 46

- Output = ?

82	**68**	**46**
45		

Convolutional layer

- Convolution operation
- Input: 5 x 5
- Filter / kernel size = 3 x 3

- Output = 3 x 3

10	**2**	**4**	**1**	**1**
0	**4**	**0**	**2**	**2**
1	**8**	**16**	**0**	**0**
8	**1**	**1**	**0**	**0**
0	**0**	**2**	**6**	**1**

*

2	**2**	**2**
1	**0**	**1**
2	**2**	**2**

=

82	**68**	**46**
45	**24**	**26**
63	**65**	**51**

Convolutional layer

Original

Sharpen

Edge Detect

"Strong" Edge Detect

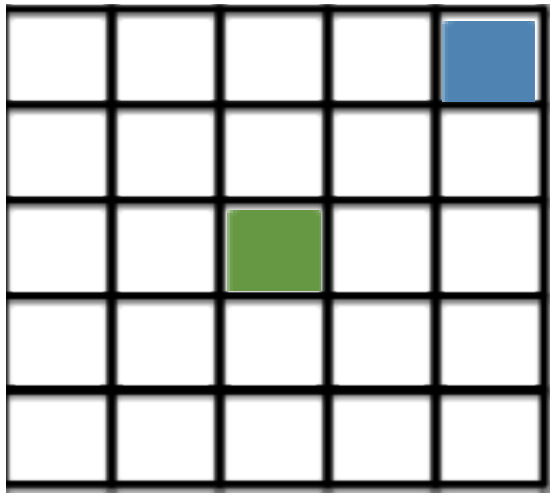
w1	w2	w3
w4	w5	w6
w7	w8	w9

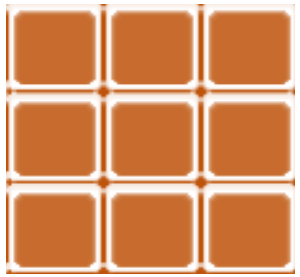
Learnable filter[1]

1. The filter weights are like the weights in Feed Forward Networks (FFN). Similar to FFN, a bias term is added after applying the filter weights.

Convolutional layer - Padding

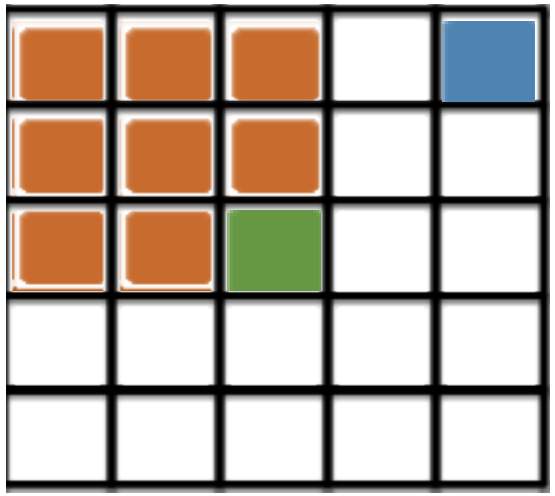
- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 0**
- **Number of times green pixel is in the filter’s receptive field: 0**





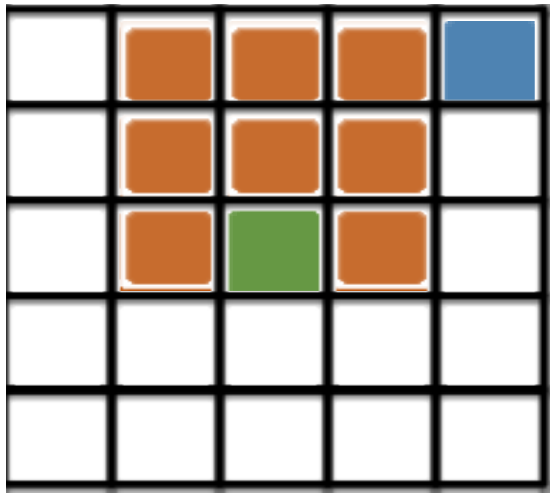
Convolutional layer - Padding

- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 0**
- **Number of times green pixel is in the filter’s receptive field: 1**



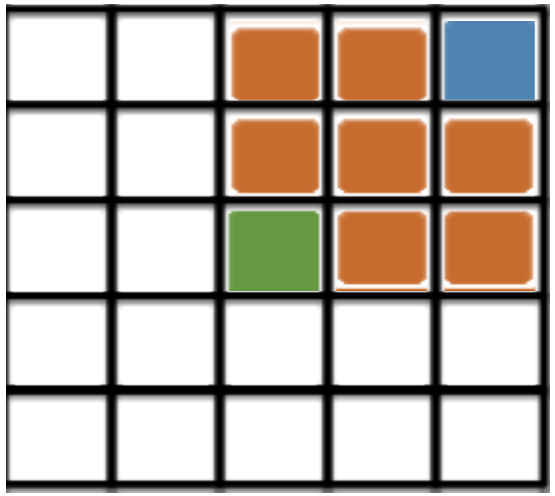
Convolutional layer - Padding

- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 0**
- **Number of times green pixel is in the filter’s receptive field: 2**



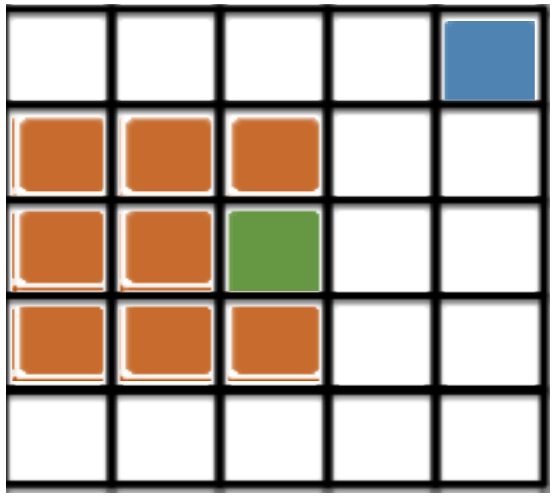
Convolutional layer - Padding

- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 1**
- **Number of times green pixel is in the filter’s receptive field: 3**



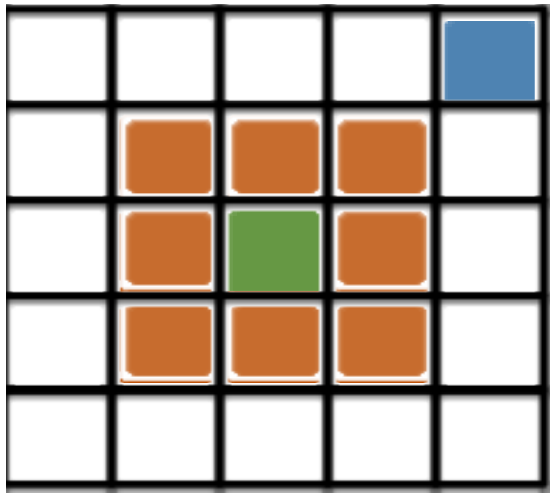
Convolutional layer - Padding

- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 1**
- **Number of times green pixel is in the filter’s receptive field: 4**



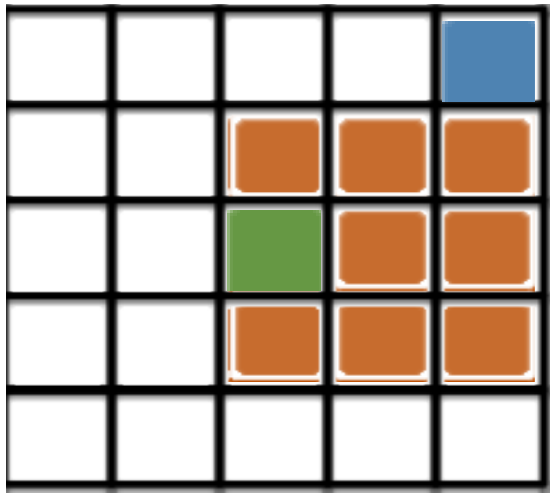
Convolutional layer - Padding

- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 1**
- **Number of times green pixel is in the filter’s receptive field: 5**



Convolutional layer - Padding

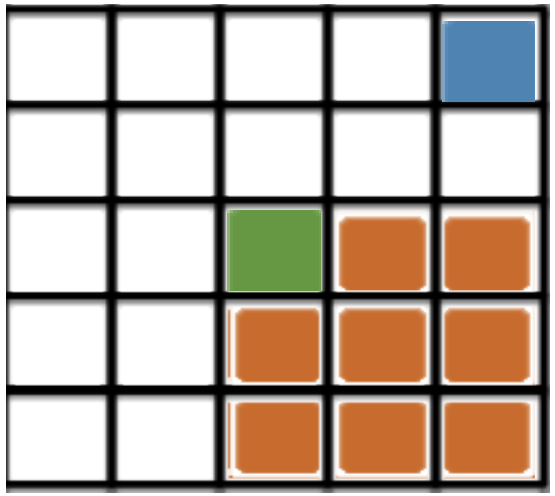
- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 1**
- **Number of times green pixel is in the filter’s receptive field: 6**



Convolutional layer - Padding

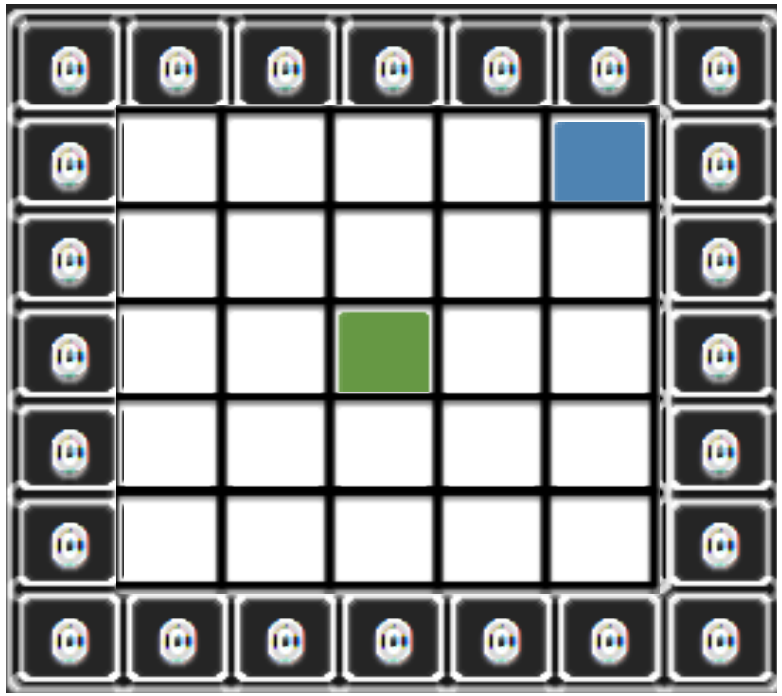
- Without Padding – corners / edges are not “seen” enough
- **Number of times blue pixel is in the filter’s receptive field: 1**
- **Number of times green pixel is in the filter’s receptive field: 9**

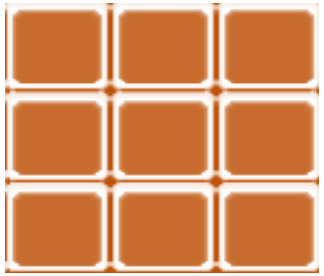
... Skip to the end



Convolutional layer - Padding

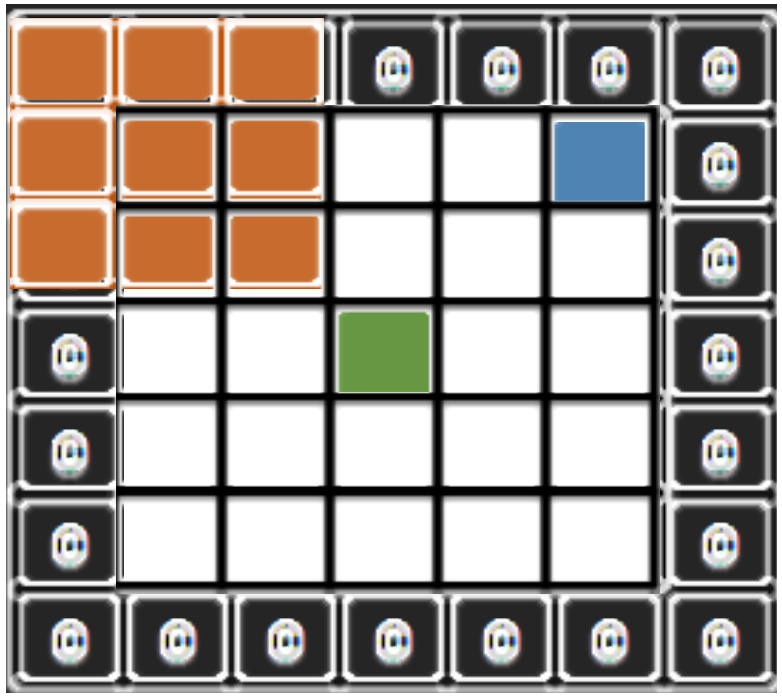
- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 0**
- **Number of times green pixel is in the filter's receptive field: 0**





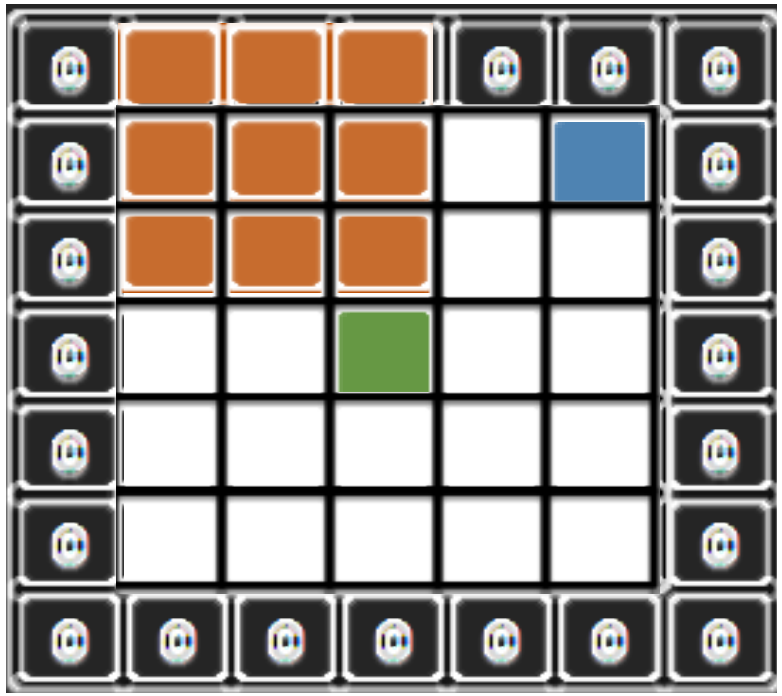
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 0**
- **Number of times green pixel is in the filter's receptive field: 0**



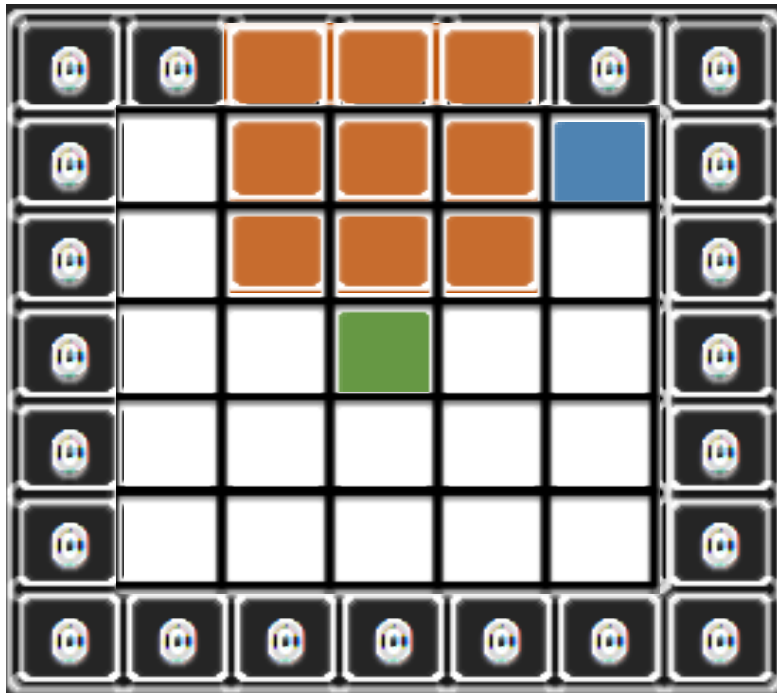
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 0**
- **Number of times green pixel is in the filter's receptive field: 0**



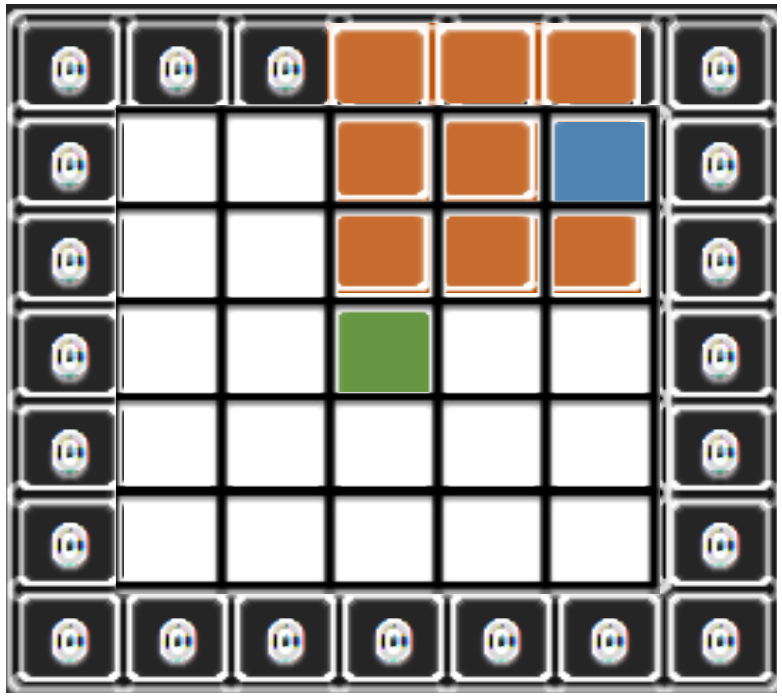
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 0**
- **Number of times green pixel is in the filter's receptive field: 0**



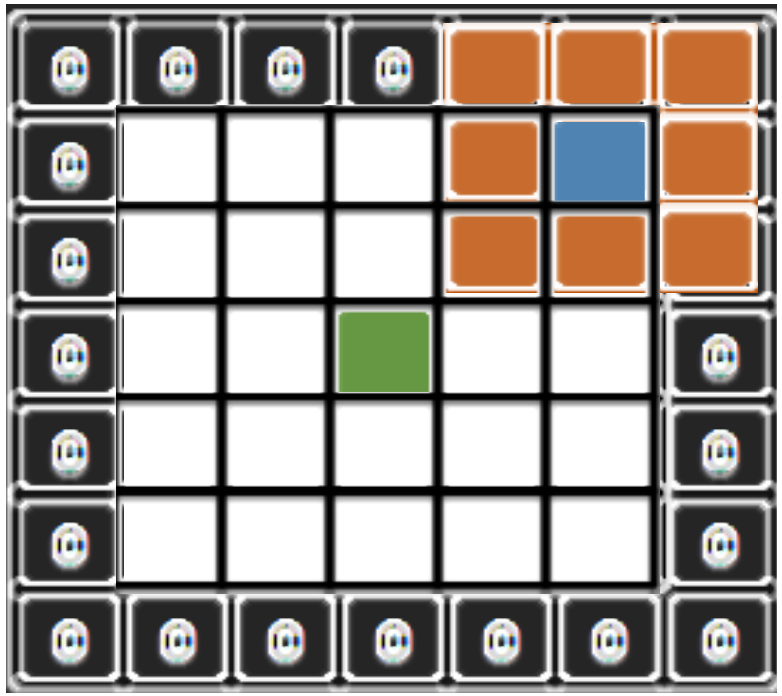
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 1**
- **Number of times green pixel is in the filter's receptive field: 0**



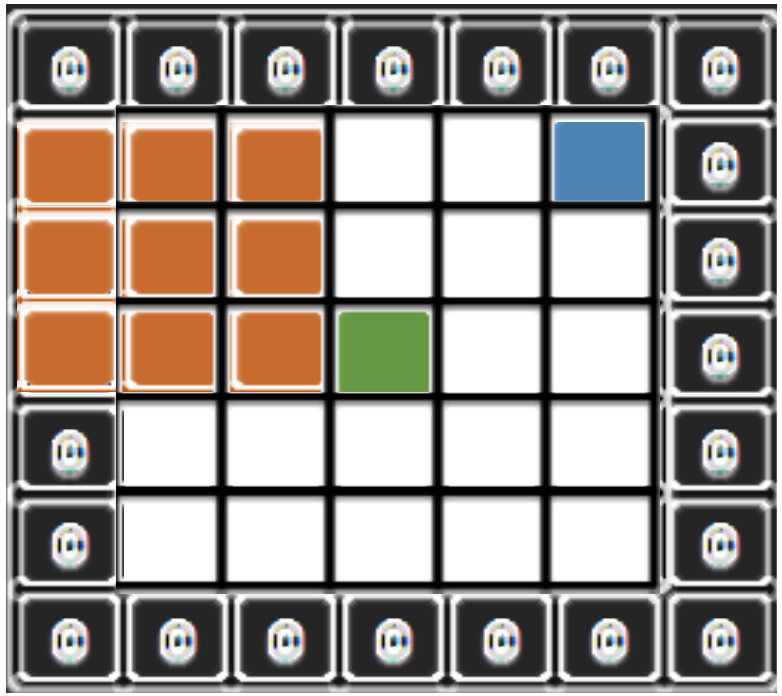
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 2**
- **Number of times green pixel is in the filter's receptive field: 0**



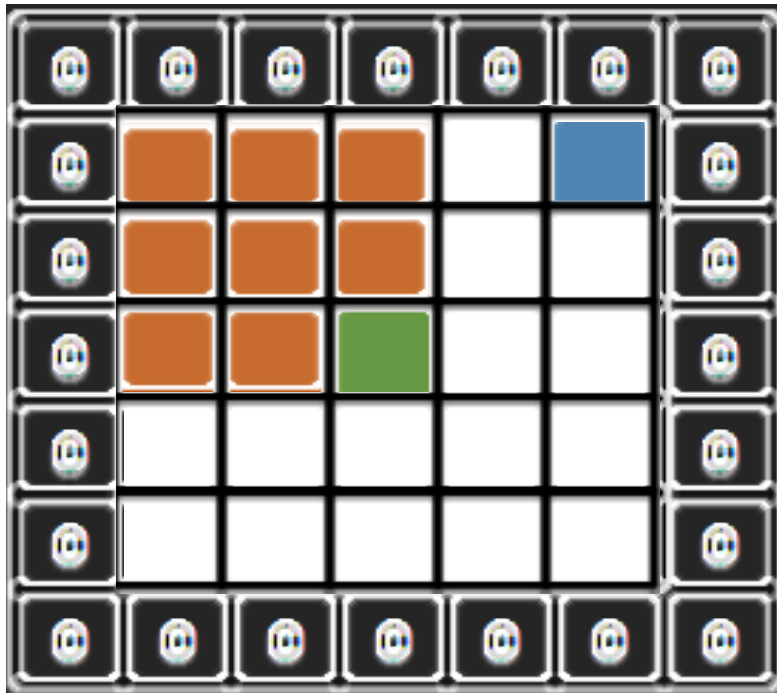
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 2**
- **Number of times green pixel is in the filter's receptive field: 0**



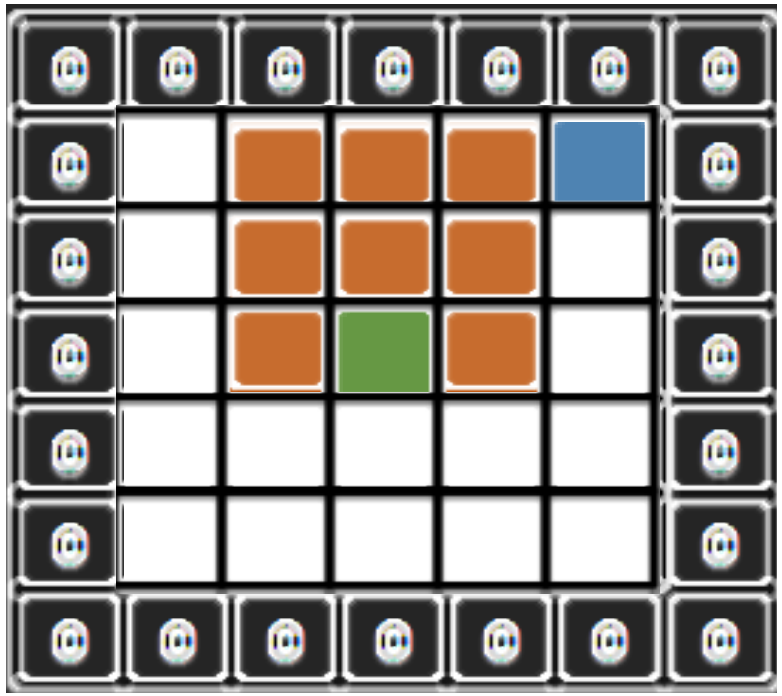
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 2**
- **Number of times green pixel is in the filter's receptive field: 1**



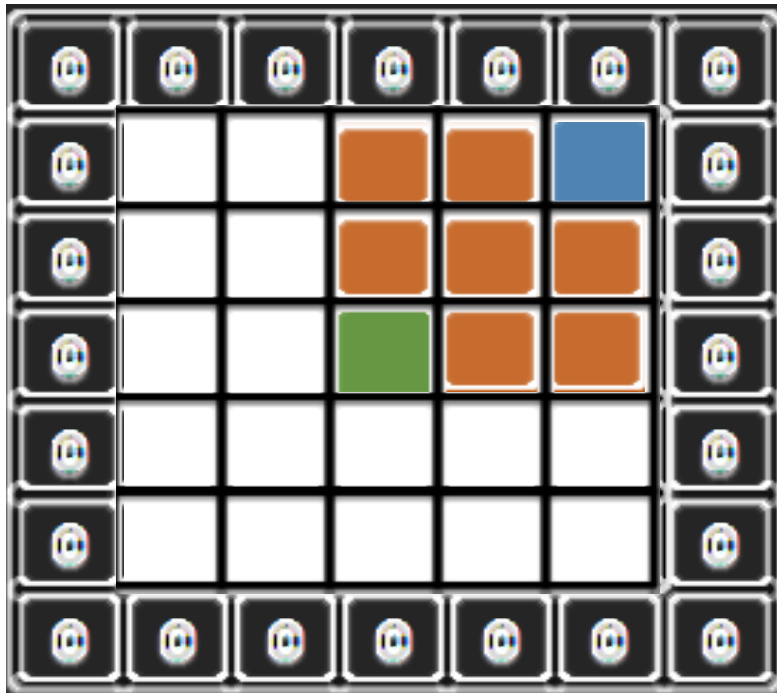
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 2**
- **Number of times green pixel is in the filter's receptive field: 2**



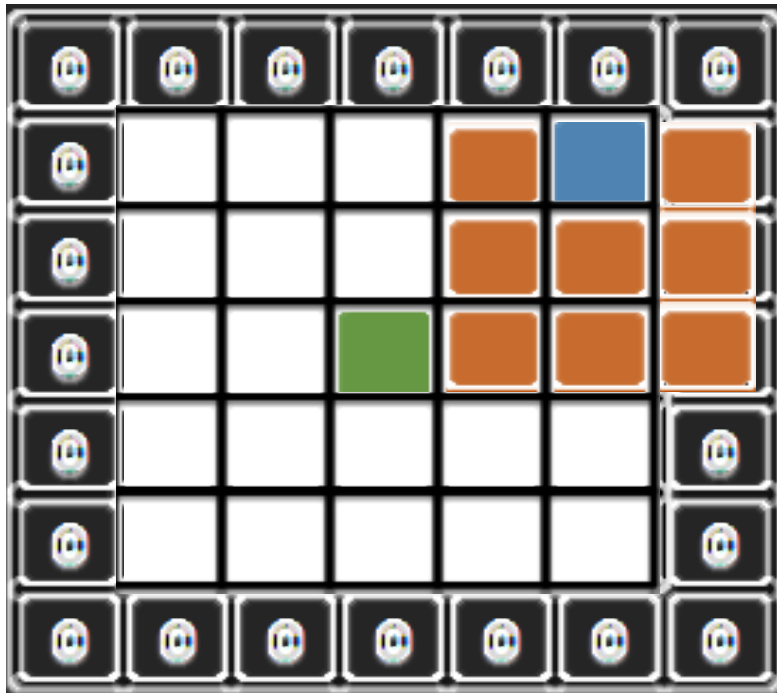
Convolutional layer - Padding

- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 3**
- **Number of times green pixel is in the filter's receptive field: 3**



Convolutional layer - Padding

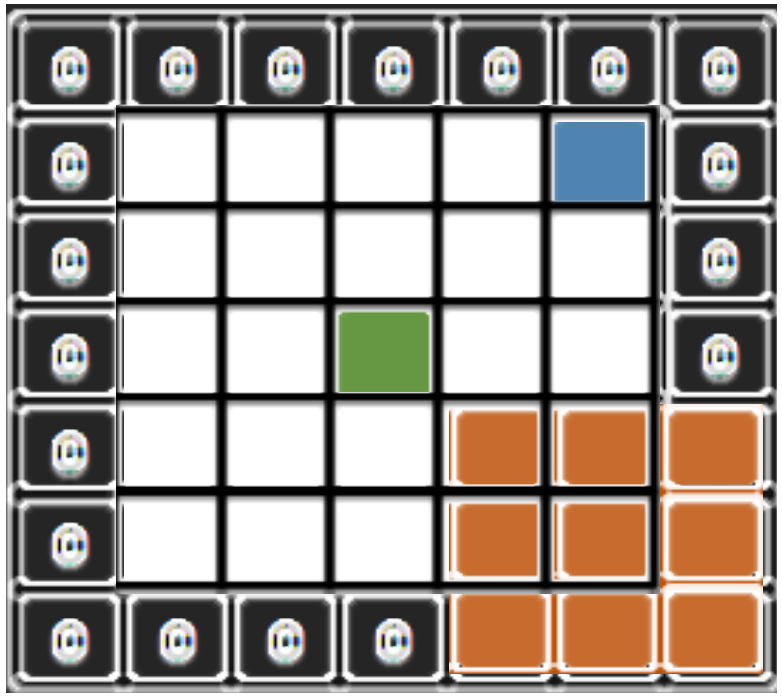
- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 4**
- **Number of times green pixel is in the filter's receptive field: 3**



Convolutional layer - Padding

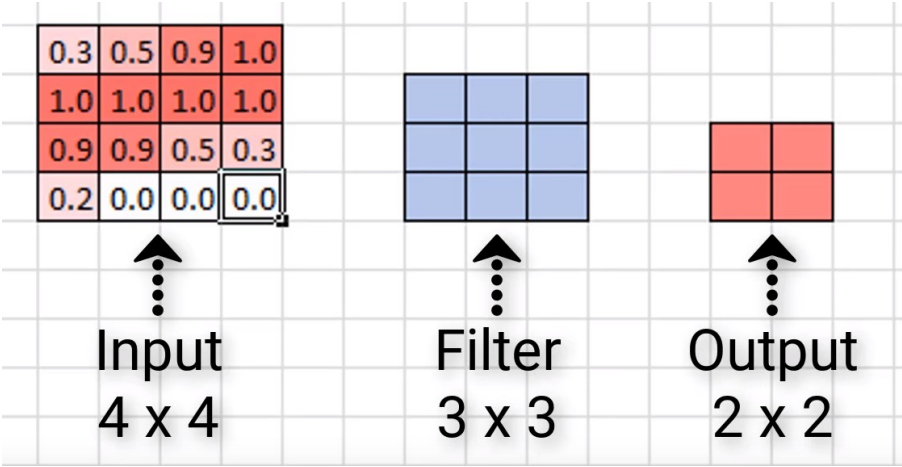
- **With Padding**
- **Number of times blue pixel is in the filter's receptive field: 4**
- **Number of times green pixel is in the filter's receptive field: 9**

... Skip to the end

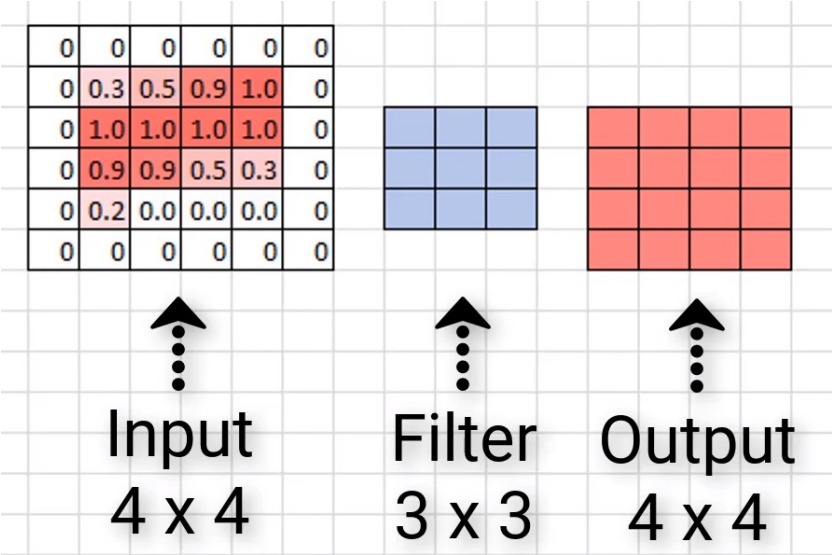


Convolutional layer - Padding

Valid Padding
p = 0

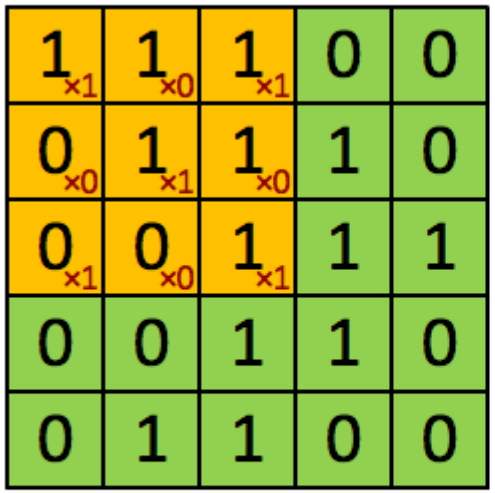


Same Padding
p ≠ 0

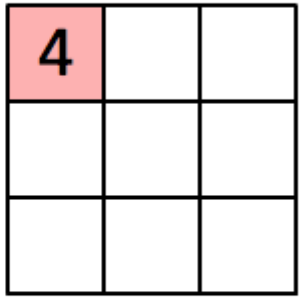


Convolutional layer - Stride

Stride
s = 1

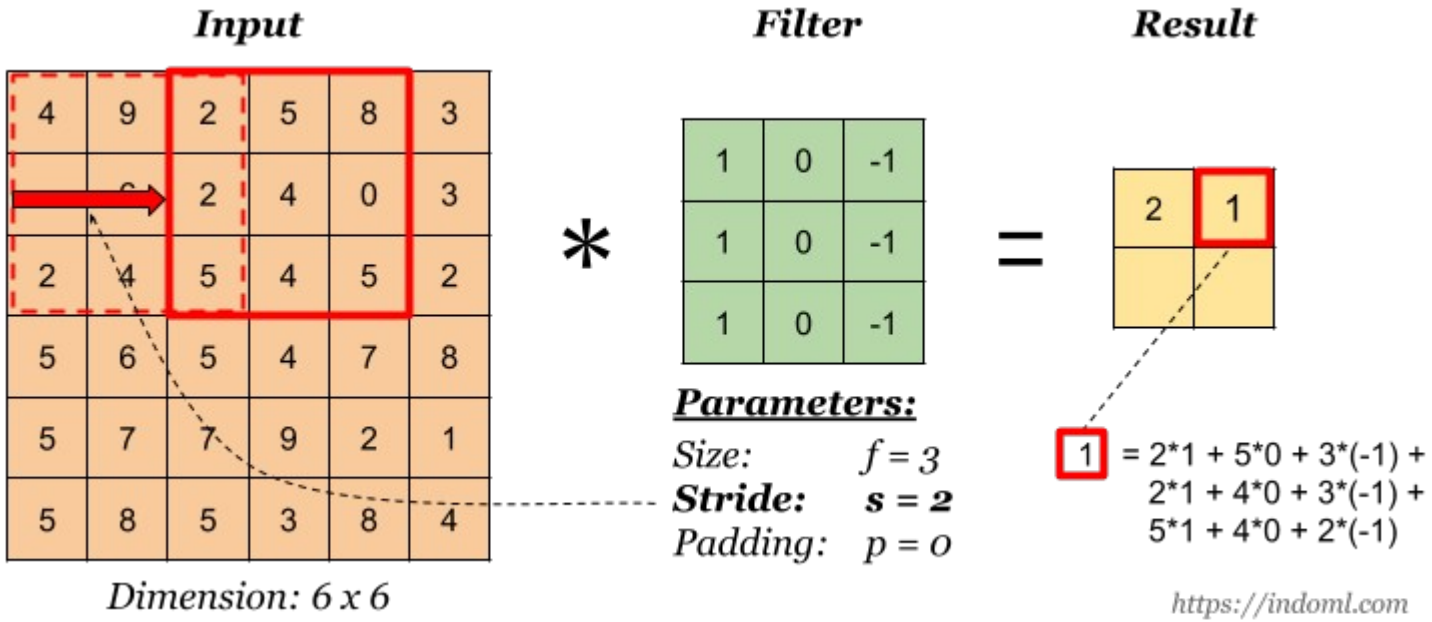


Image

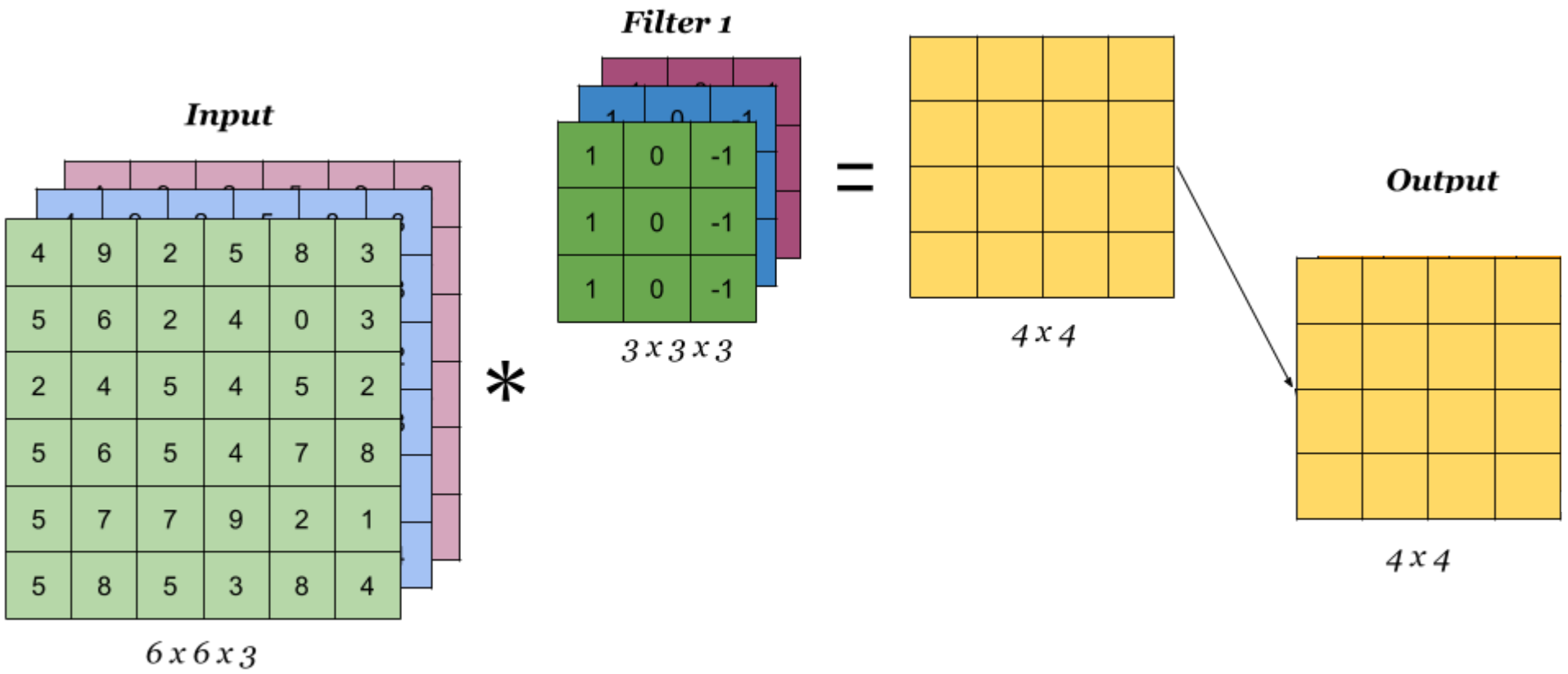


Convolved Feature

Stride
s = 2

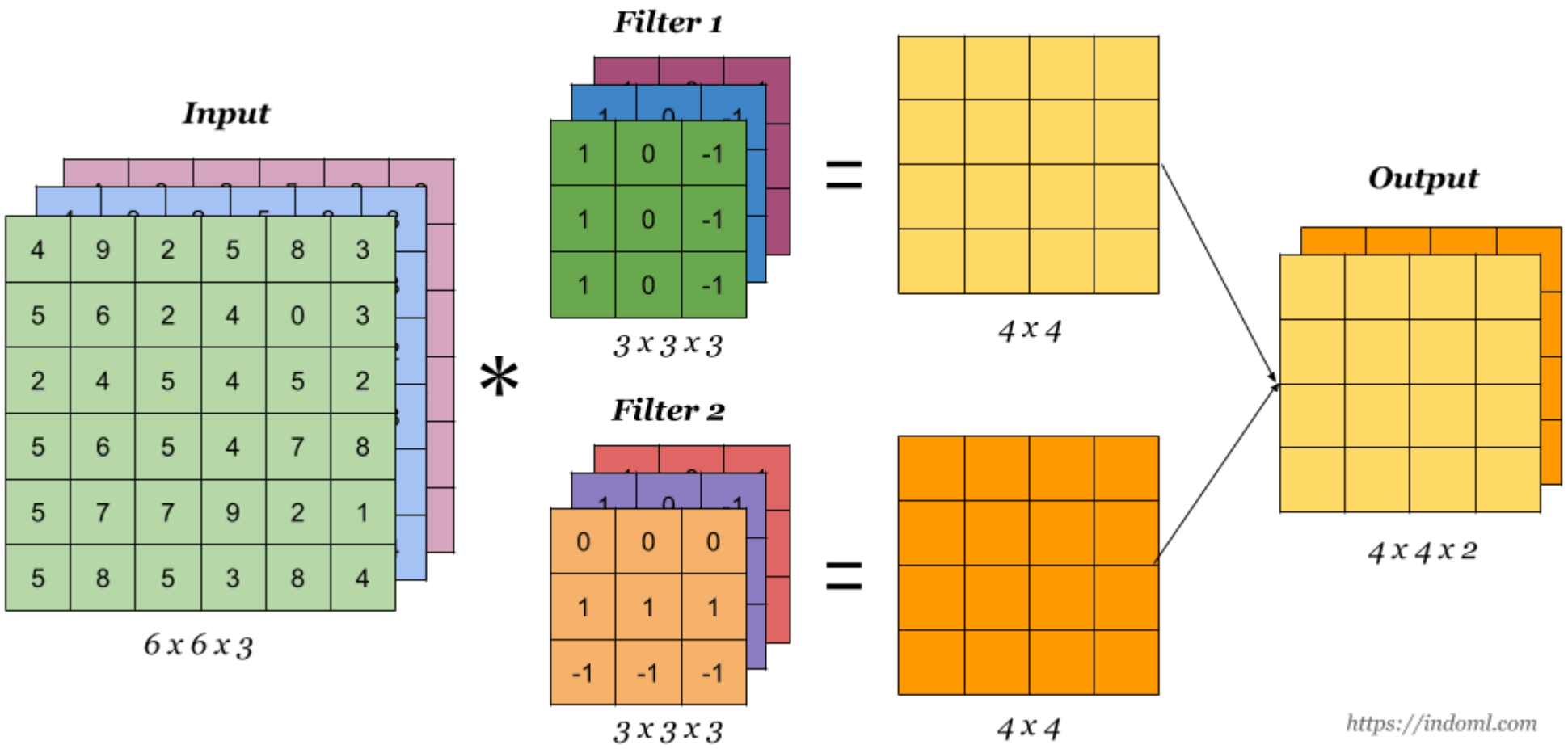


Convolutional layer – Multiple filters



https://indoml.com

Convolutional layer – Multiple filters



Convolutional Layer

Summary of convolutions

$n \times n$ image $\quad f \times f$ filter $\quad$ **m filters**

padding p $\quad$ stride s

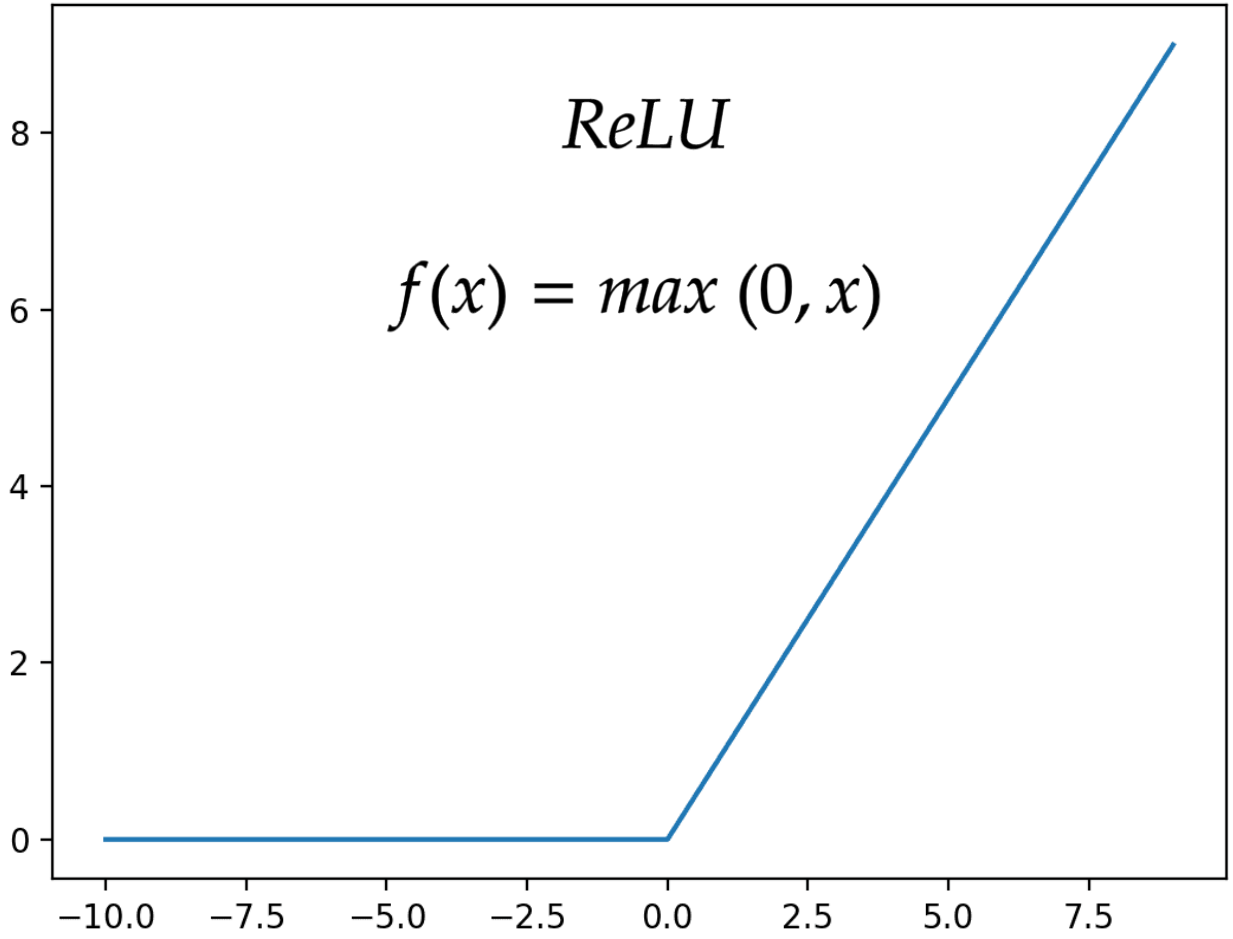
$$\left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor \times \left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor \times m$$

Example:

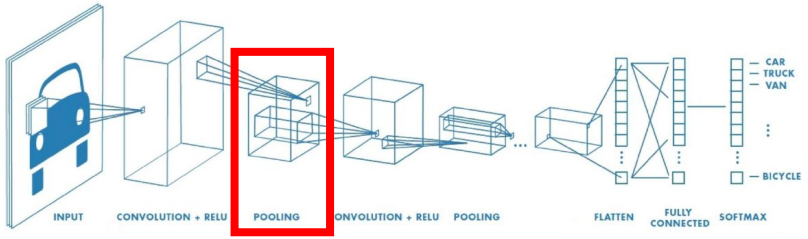
Input Image = 224x224x3 $\quad$ Filter = 3x3x3 (x32) $\quad$ Output = 222x222x32

n = 224; p = 0; s = 1; f = 3; m = 32

ReLU activation function



Pooling layer – Max Pooling



Max Pooling
Filter: 2x2
Stride: 2

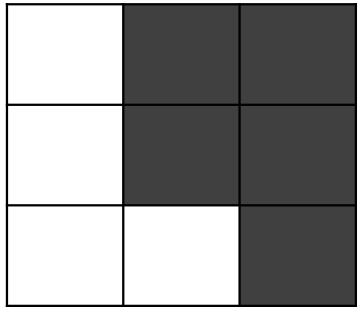
4	6	1	1
1	3	1	3
4	0	0	8
8	5	4	0

Input (4x4)

Output (2x2)

Pooling layer – Max Pooling properties

- **Translation equivariant**[1]



Input 3x3

20	0	0
20	0	0
20	20	0

$*$

Filter 2x2

2	0
0	2

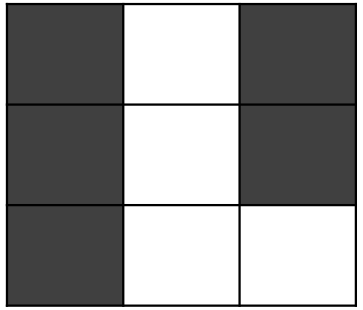
$=$

Output 2x2

40	0
80	0

Pooling 2x2 →

80



0	20	0
0	20	0
0	20	20

$*$

2	0
0	2

$=$

40	40
40	80

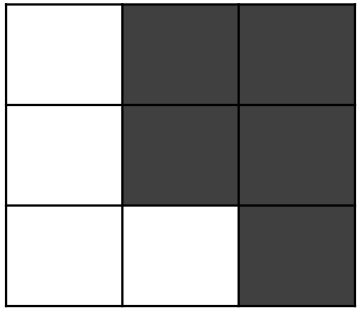
Pooling 2x2 →

80

1. Translation equivariance means that the model responds predictably to translations, while translation invariance means that the model produces the same output regardless of translations

Pooling layer – Max Pooling properties

- **Not rotation equivariant (but can be to a small degree, within the receptive field)**



Input 3x3

20	0	0
20	0	0
20	20	0

*

Filter 2x2

2	0
0	2

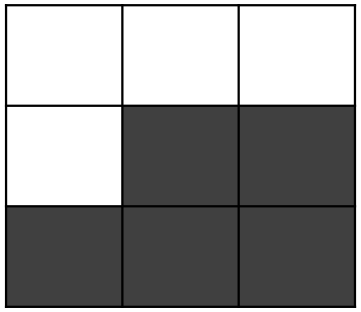
=

Output 2x2

40	0
80	0

Pooling 2x2 →

80



20	20	20
20	0	0
0	0	0

*

2	0
0	2

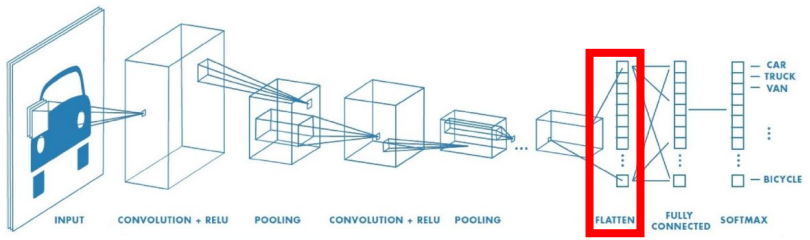
=

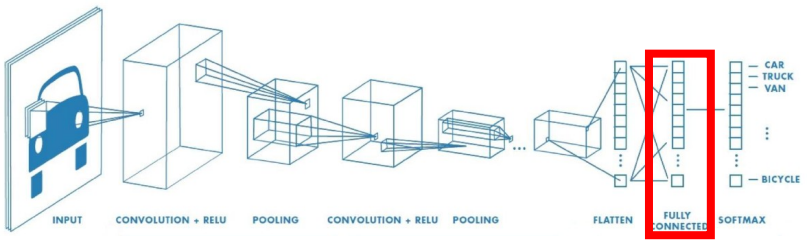
80	40
40	40

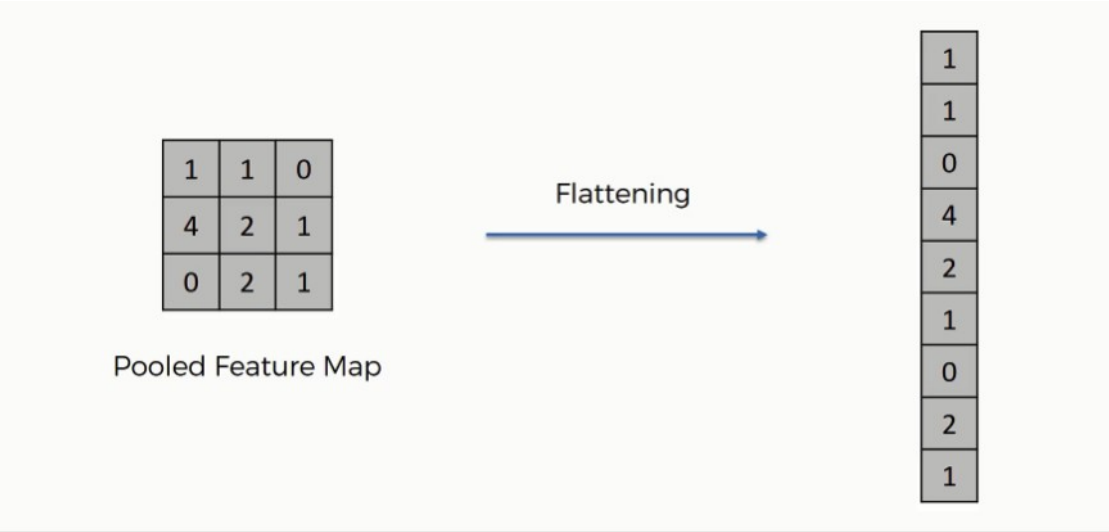
Pooling 2x2 →

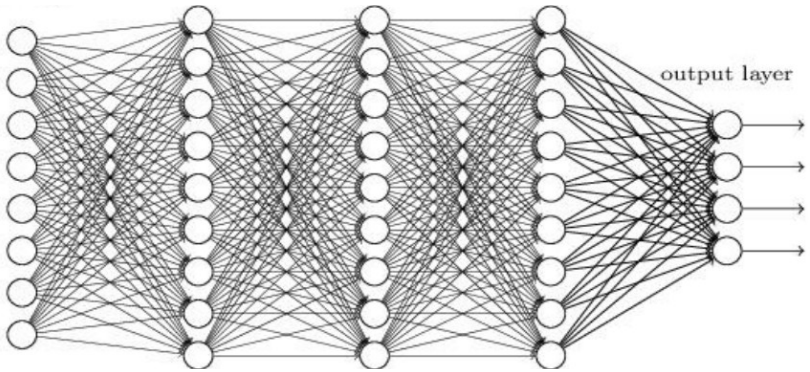
80

Flatten and Fully Connected layer





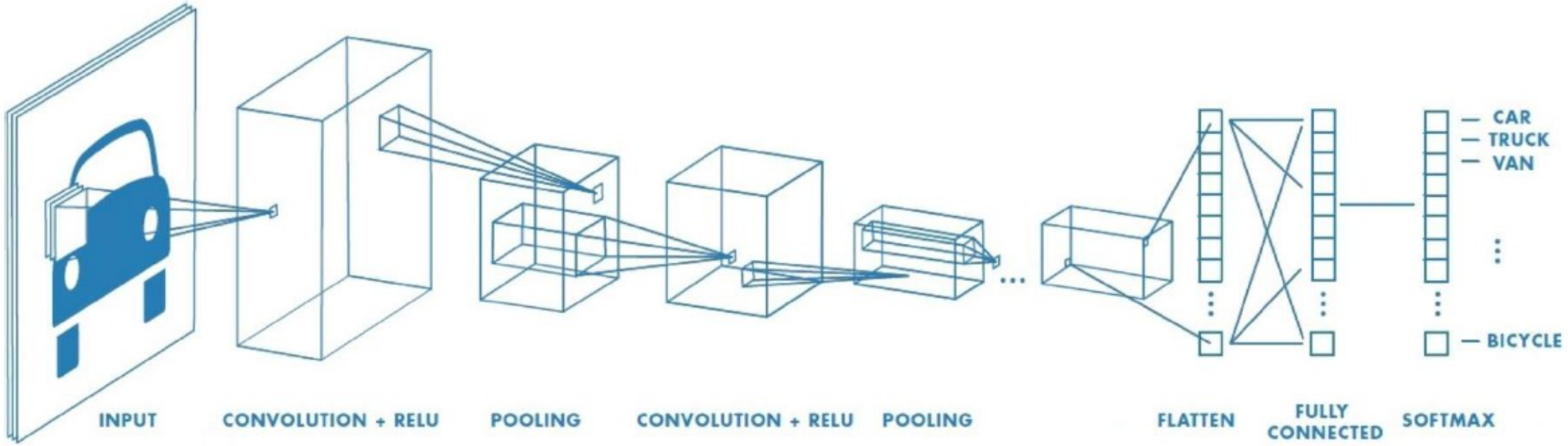




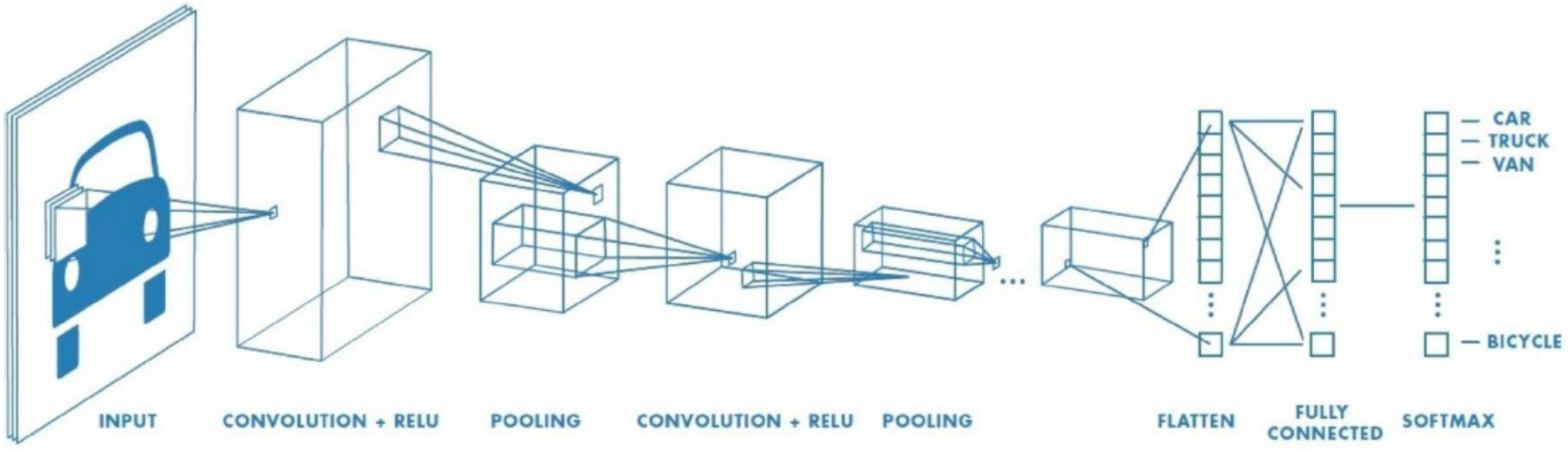
Convolutional Neural Network

Visualization:

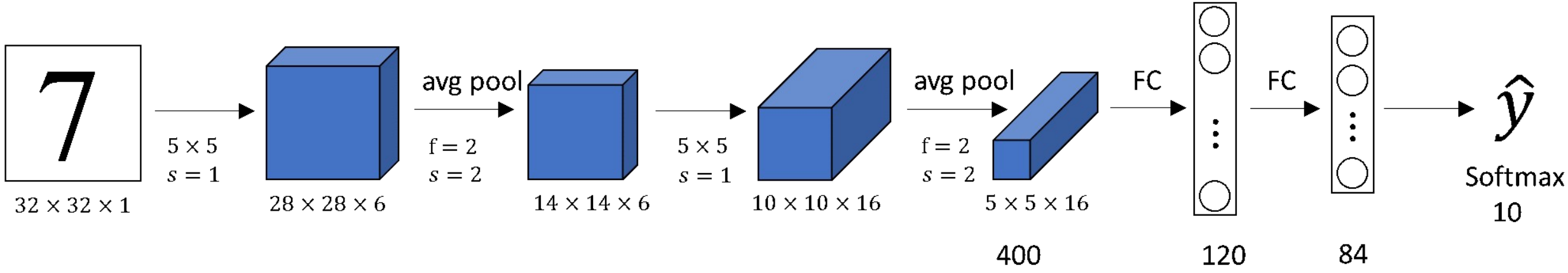
- https://poloclub.github.io/cnn-explainer/
- https://distill.pub/2017/feature-visualization/



CNN architectures



“LeNet 5” Y. Lecun et al. (1998) [3]



[3] Y. Lecun, L. Bottou, Y. Bengio and P. Haffner, "Gradient-based learning applied to document recognition," in Proceedings of the IEEE, vol. 86, no. 11, pp. 2278-2324, Nov. 1998, doi: 10.1109/5.726791.

More CNN architectures

To be continued in next lecture…

Summary

- Fully Connected Networks (Feed Forward Networks) calculate a weighted sum of the inputs
- We can add non-linear activation functions like Sigmoid to transform linear regression to logistic regression (classification)

- Classification is a supervised learning task
  - Binary: the target is 0 or 1
  - Multiclass: the target is one element out of a discrete set of elements (usually use one-hot encoding)

- Convolutional Neural Networks (CNNs) are efficient algorithms for images
  - Weight sharing
  - Capture spatial patterns
- By changing the CNN configurations, we create different architectures

Links

Backpropagtion related

- https://medium.com/binaryandmore/beginners-guide-to-deriving-and-implementing-backpropagation-e3c1a5a1e536

- (matrix form): https://sudeepraja.github.io/Neural/

Resources

Books:

- Courville, Goodfellow, Bengio: Deep Learning
  Freely available: https://www.deeplearningbook.org/
- Zhang, Aston and Lipton, Zachary C. and Li, Mu and Smola, Alexander J.: Dive into Deep Learning

  Freely available: https://d2l.ai/

Courses:

- Deep Learning specialization by Andrew NG
- https://www.coursera.org/specializations/deep-learning

That’s all for today!